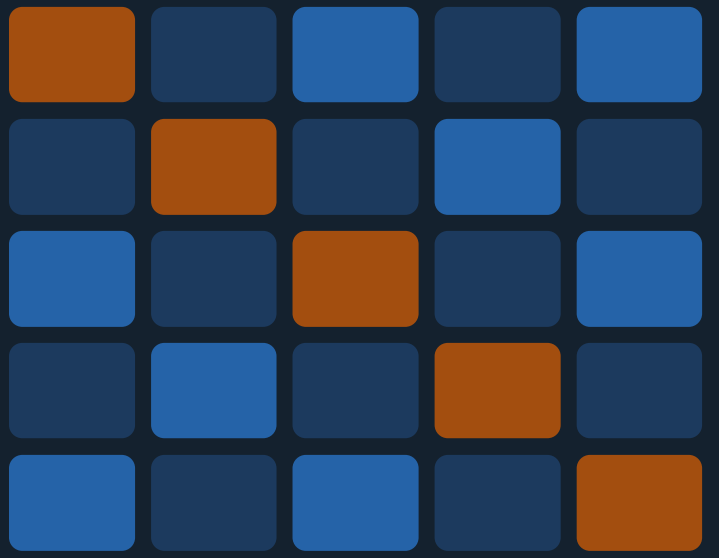
A COMPLETE PRACTITIONER HANDBOOK

Validating Machine Learning Models

From the first data split to monitoring in production: how to know, with evidence, that a model will work on data it has never seen.



BEFORE WE BEGIN

ML engineers

You train models and need a repeatable, defensible way to decide which ones ship.

Data scientists

You report results to stakeholders and want numbers that survive scrutiny.

Leads and reviewers

You sign off on models and need to know which questions to ask.

Students

You want the practical habits that textbooks mention only in passing.

KEY IDEA

The single sentence to remember

Validation is the discipline of producing **honest evidence** about how a model will behave on data it has not seen, under the conditions it will actually face.

BEFORE WE BEGIN

What you will be able to do by the end

- Choose the right data split for random, grouped, and time-ordered data
- Run and interpret cross-validation, including nested cross-validation
- Pick metrics that match the business cost of each kind of error
- Detect and prevent the many forms of data leakage
- Quantify uncertainty so a result is a range, not a single lucky number
- Find the slices where a model quietly fails
- Test fairness, robustness and calibration before launch
- Evaluate LLMs and generative systems with rubrics and judges
- Validate safely in production with shadow, canary and A/B tests
- Write a validation report a sceptical reviewer will accept

HOW TO READ THIS DECK

Every text box has a job

Throughout the tutorial, coloured boxes signal what kind of information you are looking at. Learn the code once and you can skim any slide in seconds.

KEY IDEA

The core concept of the slide. If you read nothing else, read these.

PITFALL

A mistake that teams make in real projects, often without noticing.

PRO TIP

Practical advice from experience that saves time or avoids rework.

ANALOGY

An everyday comparison that makes an abstract idea concrete.

DEFINITION

A precise meaning for a term, written in plain language.

IN THE WILD

A realistic scenario showing the idea at work in an actual project.

HOW TO READ THIS DECK

More box types you will meet

MYTH

A popular belief that sounds right but leads you astray.

REALITY

What actually holds true, and why.

RULE OF THUMB

A quick heuristic to use when you have no time to think deeply.

CHECK YOURSELF

A short question to test your understanding before moving on.

INTERVIEW ANGLE

How this topic tends to come up in technical interviews.

LESSON

The transferable takeaway from a case study.

ROADMAP

Nineteen sections, one journey

01 Why validation matters	02 Foundations of generalisation	03 Splitting your data	04 Cross-validation in depth
05 Classification metrics	06 Regression metrics	07 Metrics for other tasks	08 Imbalance, thresholds, calibration
09 Data leakage	10 Tuning and model selection	11 Uncertainty and statistics	12 Error analysis and robustness
13 Fairness and responsibility	14 Explainability as validation	15 LLMs and generative models	16 Validation in production
17 Governance and MLOps	18 End-to-end case studies	19 Toolkit and cheat sheets	

THE PREMISE

A model is only as trustworthy as the evidence you gathered about it.

Training produces a model. Validation produces the evidence. Without the second, the first is a guess with good manners.

THE MINDSET

Six principles that run through everything

1

Simulate the future

Every test should mimic how the model will meet new data in real life.

2

Protect the test set

Look at it as rarely as possible. Every peek spends some of its honesty.

3

Measure what matters

Metrics should reflect the real costs of errors, not convenience.

4

Report ranges

One number hides luck. Show the spread you would expect on a rerun.

5

Look beneath averages

Strong overall scores can hide groups where the model fails badly.

6

Keep validating

The world drifts. Validation continues after launch, not just before.

SECTION 01

01

Why validation matters

What goes wrong when models are not validated properly, and what good validation buys you.

- Cost of failure
- Training vs real performance
- Stakeholders
- What validation is not

THE CORE PROBLEM

Training scores flatter every model

A model is optimised to do well on the examples it was trained on. So its performance on those examples tells you how well it memorised, not how well it will cope with tomorrow's data.

ANALOGY

A student who has seen the exam paper in advance will score highly. The score says nothing about whether they understand the subject. Training accuracy is that pre-seen exam.

KEY IDEA

We need a separate, untouched set of examples that plays the role of the future. Everything in validation is about building and protecting that stand-in.

THE COST OF GETTING IT WRONG

When validation fails, someone pays

Financial loss

A credit model that looked excellent offline approves risky loans because the test set leaked future information.

Harm to people

A triage model under-detects illness in a group that was rare in the validation data.

Lost confidence

Stakeholders see the live numbers fall short of the promise and stop trusting the team’s results.

Wasted time

Months of engineering go into deploying a model that never had a real edge over the simple baseline.

REMEMBER

Most of these failures are not caused by bad algorithms. They come from evidence that looked solid but answered the wrong question.

A FAMILIAR STORY

The offline–online gap

1. **Great offline score**
   The team reports a big improvement on the held-out set.
2. **Launch**
   The model goes live with an announcement and a dashboard.
3. **Disappointing live results**
   Business metrics barely move, or get worse.
4. **Investigation**
   Leakage, a split that ignored time, or a metric that did not match the goal is found.
5. **Rework**
   The fix takes longer than doing validation properly in the first place.

IN THE WILD

A churn model scored superbly in testing. Later it emerged that one feature was set by the retention team **after** a customer had already announced they were leaving. Offline, it was a near-perfect signal. Live, it simply did not exist yet at prediction time.

WHAT VALIDATION GIVES YOU

Four things good validation buys

01

An honest estimate

How well the model will do on new data, with a realistic range around it.

02

A fair comparison

Confidence that model B is genuinely better than model A, not just luckier.

03

Known weaknesses

A map of where the model fails, so you can fix it, guard it or warn users.

04

A defensible decision

Evidence you can show to a reviewer, regulator or customer.

CLEARING UP TERMS

Validation, evaluation, verification, testing

Term	What it usually means	Question it answers
Evaluation	Measuring a model’s performance with chosen metrics	How good is it?
Validation	Gathering evidence the model is fit for its intended use	Should we trust it for this job?
Verification	Checking the system was built as specified	Did we build it right?
Testing	Software checks on code, data and pipelines	Does it behave as designed?
Validation set	Data held out for tuning and model choice	Which version do we pick?
Test set	Data held out for the final, one-off estimate	How will the chosen model do?

⚠ PITFALL

Some fields swap the names “validation set” and “test set”. Always state which set was used for what in any report, so nobody has to guess.

CLEARING UP MISCONCEPTIONS

What validation is not

MYTH

Validation is a single step at the end of a project.

REALITY

It shapes the project from the first split to post-launch monitoring.

MYTH

A high accuracy number means the model is ready.

REALITY

One number can hide imbalance, leakage or failing groups.

MYTH

If cross-validation looks good, the model is safe.

REALITY

Cross-validation is only as good as the way folds mirror real use.

THE BIGGER PICTURE

Validation sits across the whole lifecycle

1 → 2 → 3 → 4 → 5 → 6

Frame

Define the decision, users and cost of errors.

Split

Design held-out data that mimics the future.

Develop

Train and tune using validation data only.

Confirm

Final test, slices, fairness and robustness.

Deploy

Shadow, canary and online experiments.

Monitor

Track drift and live performance continuously.

PRO TIP

Write a one-page **validation plan** before you train anything: the split, the metrics, the baselines and the pass criteria. It stops you moving the goalposts once results come in.

WHO CARES ABOUT THE RESULT

Different stakeholders, different questions

Stakeholder	Their real question	Evidence they need
Product owner	Will this improve the outcome we care about?	Business-aligned metric, online test
Engineering	Will it be reliable and fast enough?	Latency, failure modes, robustness tests
Risk and compliance	Can we defend it if challenged?	Documentation, fairness checks, audit trail
End users	Will it treat me fairly and correctly?	Slice performance, explanations
Leadership	Is it worth the cost and risk?	Gain over baseline with uncertainty

KEY IDEA

A validation report should answer every row, in the language of each audience.

SECTION 1 RECAP

Why validation matters

REMEMBER

Takeaways

- Training scores measure memory, not skill
- Failures usually come from flawed evidence, not flawed algorithms
- Validation spans the whole lifecycle

CHECK YOURSELF

Your team reports 98% accuracy on the data the model was trained on. What is the first question you should ask?

ANSWER

“What is the score on data the model never saw during training or tuning, and how was that data chosen?”

SECTION 02

02

Foundations of generalisation

The ideas that explain why models fail on new data: overfitting, underfitting, bias, variance and the assumptions behind every test.

Generalisation | Overfitting | Underfitting | Bias and variance | Learning curves | Baselines

THE GOAL

Generalisation is the whole point

DEFINITION

Generalisation

Generalisation is a model's ability to perform well on new examples drawn from the same situation it will be used in, not just on the examples it learned from.

IN PLAIN WORDS

We want a model that has learned the underlying pattern, not one that has memorised the particular examples it happened to see.

ANALOGY

A good cook understands why recipes work and can improvise with new ingredients. A poor one can only reproduce the dishes they practised.

FAILURE MODE ONE

Overfitting: learning the noise

An overfit model has captured quirks, coincidences and noise in the training data as if they were real patterns. It looks brilliant on familiar data and stumbles on anything new.

WARNING SIGNS

- Training score far better than validation score
- Performance jumps around between folds or reruns
- Very complex model on a small dataset
- Predictions hinge on oddly specific features

ANALOGY

Memorising that every photo of a wolf in your training set has snow in it, then calling any dog in the snow a wolf.

THE FIX

- More or better data
- Simpler model or stronger regularisation
- Early stopping
- Fewer, more meaningful features

FAILURE MODE TWO

Underfitting: missing the pattern

An underfit model is too simple, too constrained, or too poorly fed to capture the real structure. It does badly everywhere, including on its own training data.

WARNING SIGNS

- Training and validation scores are both poor
- Scores barely improve with more data
- Errors follow a clear, systematic pattern
- Model barely beats a trivial baseline

ANALOGY

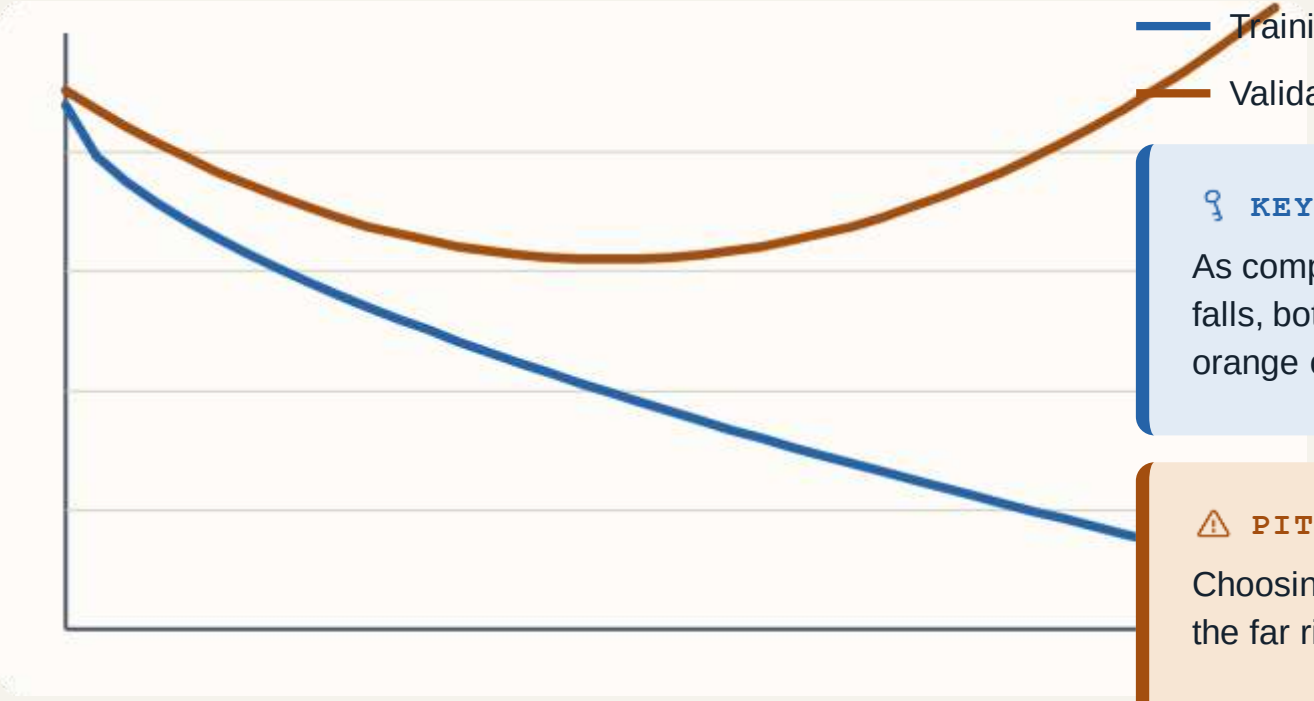
Trying to describe a winding road using only a straight ruler. No amount of extra practice will make the ruler bend.

THE FIX

- Richer features or representations
- More flexible model
- Less regularisation
- Train longer or tune better

THE CLASSIC PICTURE

Complexity versus error



Across: model complexity · Up: error

KEY IDEA

As complexity grows, training error keeps falling. Validation error falls, bottoms out, then rises. The sweet spot is the bottom of the orange curve.

PITFALL

Choosing the model with the lowest training error always picks the far right of this chart: the most overfit option.

TWO SOURCES OF ERROR

Bias and variance, without the maths

Bias

Error from wrong assumptions. A high-bias model is consistently off in the same direction, no matter which sample it learns from. It underfits.

Variance

Error from sensitivity to the particular sample. A high-variance model changes a lot when trained on a slightly different dataset. It overfits.

Irreducible noise

Randomness in the world that no model can predict. It sets a floor on achievable error.

ANALOGY

Think of archers. A biased archer hits a tight cluster in the wrong place. A high-variance archer scatters arrows all around the target. A good archer is both accurate and consistent.

TWO SOURCES OF ERROR

Reading bias and variance from your results

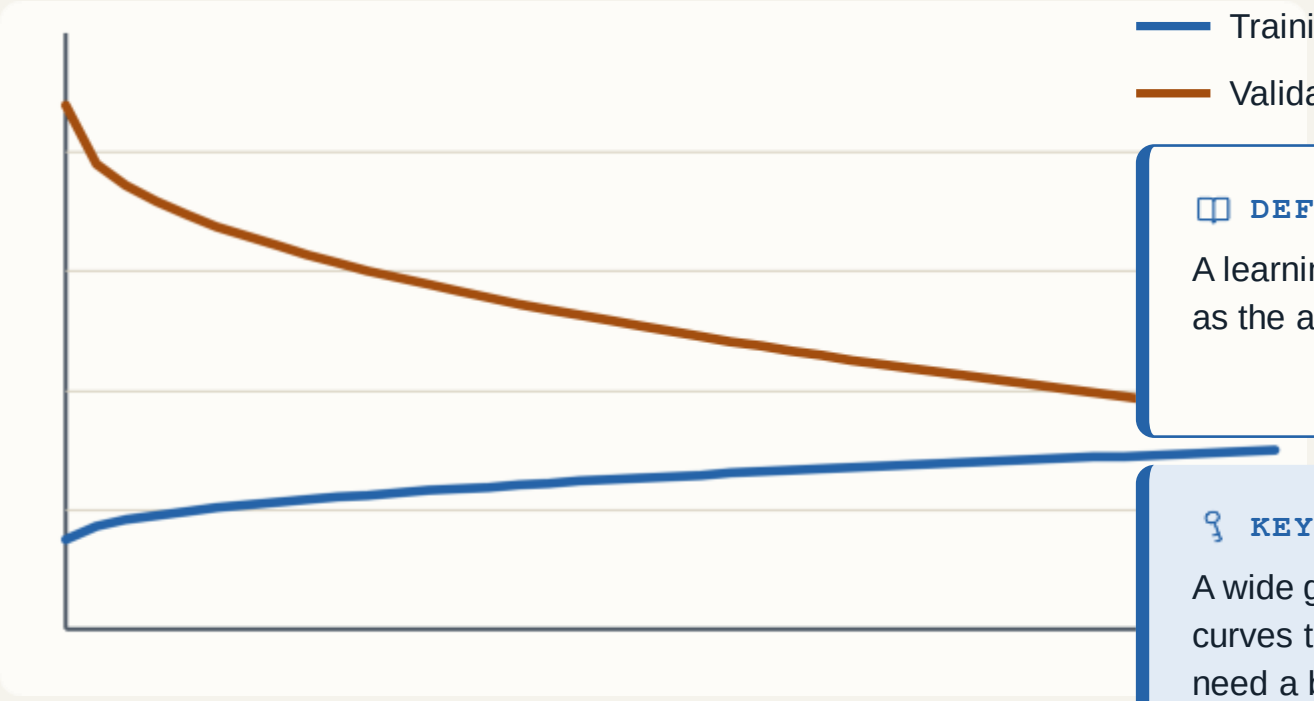
What you see	Likely problem	What to try
Poor training score, poor validation score	High bias (underfitting)	More capacity, better features
Great training score, poor validation score	High variance (overfitting)	More data, regularisation, simpler model
Good training score, good validation score	Healthy fit	Confirm on test and slices
Validation score varies wildly across folds	High variance or small data	More data, repeated CV, simpler model
Both scores fine, live performance poor	Not bias or variance: a mismatch	Check leakage and data drift

PRO TIP

The last row matters most in practice. When offline and online disagree, suspect the **validation design**, not the model.

DIAGNOSTIC TOOL

Learning curves



Across: training set size · Up: error

DEFINITION

A learning curve plots performance on training and validation data as the amount of training data increases.

KEY IDEA

A wide gap that is still closing means more data will help. Two curves that have met at a poor level mean more data will not; you need a better model or features.

DIAGNOSTIC TOOL

Reading learning curves

Gap still closing

Variance problem. Collecting more data is likely to pay off.

Converged high

Bias problem. More data will not help much; change the model or features.

Converged low

Healthy. Further gains will need new information, not more of the same.

Erratic

Too little data per point or unstable training. Repeat with several seeds.

PRO TIP

Learning curves are cheap insurance before a costly data-collection project. They tell you whether buying more labels is likely to be worth it.

DIAGNOSTIC TOOL

Validation curves for one setting

DEFINITION

A validation curve shows training and validation scores as you vary a single hyperparameter, such as tree depth or regularisation strength.

WORKED EXAMPLE

For tree depth: shallow trees score poorly on both sets; very deep trees score perfectly on training but worse on validation. The depth where validation peaks is your candidate.

PITFALL

Picking the peak of a validation curve is itself a form of fitting. Once you choose based on it, that validation score becomes optimistic. You still need an untouched test set or nested cross-validation.

THE HIDDEN ASSUMPTION

Same conditions in, same conditions out

Every standard validation method assumes that the data you test on comes from the same process as the data the model will see in use. When that assumption breaks, even perfect validation gives the wrong answer.

DEFINITION

Covariate shift: the inputs change, but the relationship to the outcome stays the same.

DEFINITION

Label shift: how common each outcome is changes, for example fraud rises sharply.

DEFINITION

Concept drift: the relationship itself changes, for example customer behaviour after a price rise.

KEY IDEA

Validation design is mostly about making the held-out data resemble the deployment world as closely as possible.

ALWAYS COMPARE

Baselines: the most underrated validation tool

A score means nothing on its own. Is 85% good? Only if the simplest sensible approach gets much less.

RULE OF THUMB

No model ships until it beats a trivial baseline **and** the best simple baseline, by a margin larger than the uncertainty in the scores.

Baseline	Example
Majority class	Always predict the most common label
Mean or median	Always predict the average value
Last value	Tomorrow equals today in forecasting
Seasonal naive	Same as this day last week
Simple rule	The heuristic the business already uses
Linear model	A plain regression on key features

ALWAYS COMPARE

Why baselines catch so many problems

IN THE WILD

A model scores 99% accuracy detecting a rare defect. The majority-class baseline also scores 99%, because 99% of parts have no defect. The model learned nothing useful.

IN THE WILD

A demand forecast looks sophisticated, but “same as last week” is only slightly worse. The complex pipeline adds cost and fragility for a tiny gain.

IN THE WILD

A model beats the baseline by a wide margin. Suspiciously wide. Investigation finds a leaked identifier. Baselines expose too-good-to-be-true results, too.

SECTION 2 RECAP

Foundations of generalisation

REMEMBER

Takeaways

- Overfitting learns noise; underfitting misses signal
- Bias is consistent error; variance is instability
- Learning curves tell you whether more data will help
- Every test assumes the future looks like the held-out data
- Always beat sensible baselines

CHECK YOURSELF

Training error is very low and validation error is high, but the gap shrinks as you add data. What is happening and what should you do?

ANSWER

Overfitting driven by variance. More data should help; meanwhile try regularisation or a simpler model.

SECTION 03

03

Splitting your data

How to carve data into training, validation and test sets so that the held-out parts genuinely stand in for the future.

Hold-out | Three-way split | Stratification | Group splits | Time splits | Split sizes

THE BASIC MOVE

The hold-out split



DEFINITION

Set aside a portion of the data before any modelling. Train on the rest. Score the model once on the held-out portion to estimate performance on unseen data.

USE IT WHEN

You have plenty of data, compute is limited, or you need a quick first estimate.

AVOID IT WHEN

Data is small: one random split can be lucky or unlucky, and the estimate will be noisy.

THE STANDARD DESIGN

Train, validation and test



DEFINITION

Training set: the model learns its internal settings from these examples.

DEFINITION

Validation set: used to compare models, tune hyperparameters and pick thresholds.

DEFINITION

Test set: touched once, at the end, to estimate how the chosen model will perform.

RULE OF THUMB

If a decision was made using a dataset, that dataset can no longer give an unbiased estimate of the result of that decision.

WHY THREE SETS

Every look at data spends its honesty

Each time you choose something based on a dataset, the model’s score on that dataset becomes a little optimistic, because you selected the option that happened to do well there.

ANALOGY

If you roll twenty dice and keep the highest, the score you kept overstates how good a typical roll is. Tuning on validation data is choosing the best roll.

KEY IDEA

The validation score of your final model is biased upwards. The test set exists to give one fresh, unbiased reading after all choices are frozen.

PITFALL

Teams that “just check” the test set after each experiment have silently turned it into a second validation set. Its estimate is then no longer trustworthy.

PROTECTING THE TEST SET

House rules for the test set

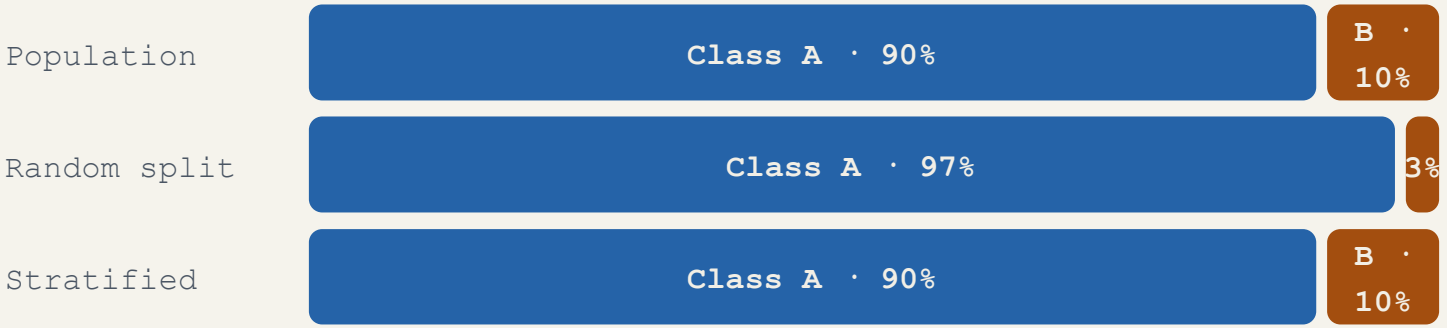
- Create it first, before exploring the data in depth
- Store it separately, with restricted access if possible
- Never use it for feature selection, scaling or tuning
- Log every time it is evaluated, and by whom
- Evaluate only frozen, final candidates
- If it has been overused, retire it and collect a fresh one

PRO TIP

Some teams hash the test-set IDs and add a check to their pipeline that refuses to train if any test ID appears in the training data. Cheap, automatic and surprisingly effective.

KEEPING PROPORTIONS

Stratified splitting



A random split of imbalanced data can leave the held-out set with too few rare cases. Stratification keeps class proportions the same in every part.

USE IT WHEN

Classification with imbalanced classes; small datasets; multi-class problems with rare labels.

PRO TIP

You can stratify on more than the label: for example on region or customer type, to keep important subgroups represented in every split.

HIDDEN DUPLICATES

Group splits: keep related rows together

When several rows come from the same source, a random split puts some in training and some in testing. The model then recognises the source rather than learning the task.

Same person

Multiple visits, scans or transactions from one patient or customer.

Same object

Several photos of the same product, or readings from the same machine.

Same place

Neighbouring map tiles or houses on the same street.

Same document

Sentences or chunks cut from one article or contract.

THE FIX

Split by a **group identifier** so every row from one group lands in the same part. Then the test genuinely asks: does this work for a group we have never seen?

HIDDEN DUPLICATES

What a group leak looks like

IN THE WILD

Medical imaging

A skin-lesion classifier scores superbly. Patients often had several photos taken in one visit, and the split was random. Many test images had a near-twin in training. When re-split by patient, performance dropped sharply.

IN THE WILD

Voice

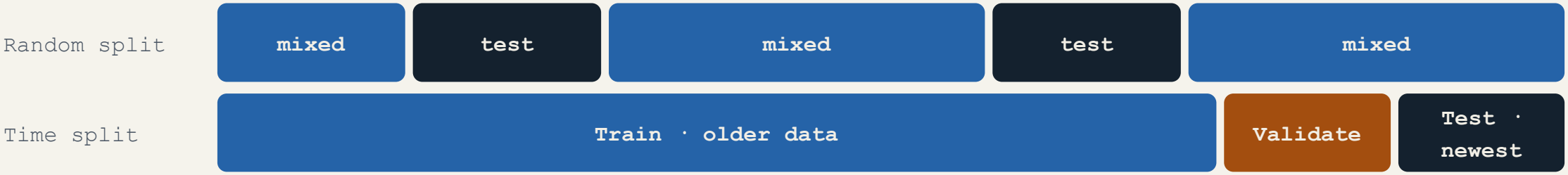
A speaker-recognition model is tested on new recordings of the same speakers it trained on. It is then deployed for new customers, where it struggles. The test asked the wrong question.

ASK YOURSELF

Ask: “In deployment, will the model meet entities it has already seen, or brand-new ones?” Your split must answer the same question.

TIME MATTERS

Time-based splits



KEY IDEA

If the model will predict the future from the past, the test must also be in the future relative to the training data. A random split lets the model peek at tomorrow.

PITFALL

Random splits on time-ordered data are one of the most common causes of models that shine offline and fail in production.

TIME MATTERS

Gaps and embargoes



DEFINITION

A **gap** (or embargo) is a stretch of time left unused between training and test periods.

WHY IT MATTERS

Labels often take time to mature, and features computed over rolling windows can overlap the boundary. The gap stops information from bleeding across.

WORKED EXAMPLE

Predicting whether a loan defaults within 12 months? Training labels near the cut-off are not known yet at prediction time. Leave a gap at least as long as the outcome window.

CHOOSING SIZES

How big should each part be?

Situation	Typical approach	Reasoning
Small data (hundreds of rows)	Cross-validation, small or no fixed test set	Every row is precious; reuse it carefully
Medium (thousands)	Around 60/20/20 or 70/15/15	Enough for stable estimates in each part
Large (millions)	98/1/1 can be plenty	A small fraction is still a big sample
Rare outcomes	Size test set by positive count	Hundreds of positives matter more than total rows
Many subgroups	Ensure each group has enough test rows	Slice results need their own sample size

RULE OF THUMB

Size the test set by how precisely you need to know the answer, not by a fixed percentage. A wider range of uncertainty is acceptable for a prototype, not for a medical device.

BEFORE YOU SPLIT

Questions to answer first

ASK YOURSELF

Will the model see new **people**, **products**, or **places** in deployment? Then split by that group.

ASK YOURSELF

Will the model predict **forward in time**? Then split by time, with a gap.

ASK YOURSELF

Are some outcomes **rare**? Then stratify and check positive counts.

ASK YOURSELF

Could there be **duplicates** or near-duplicates? Then deduplicate before splitting.

ASK YOURSELF

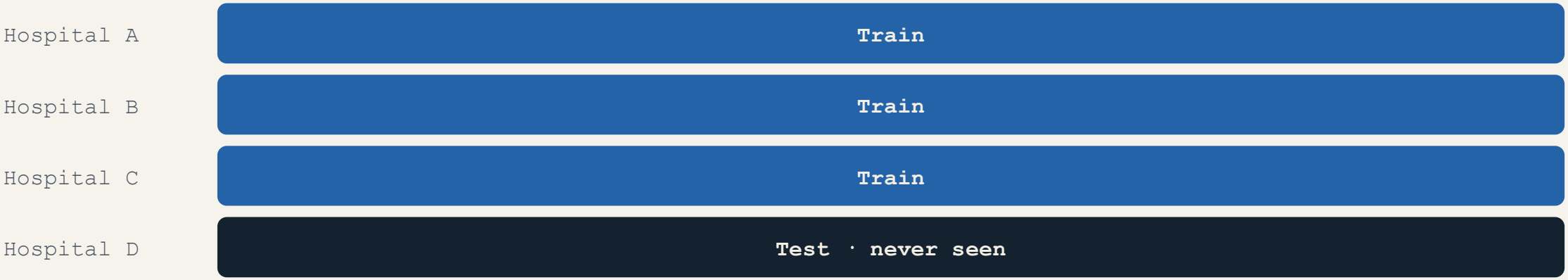
Will deployment data come from a **new site** or source? Then hold out a whole site.

ASK YOURSELF

Are labels **delayed**? Then leave a gap and use only matured labels.

REAL-WORLD SPLITS

Holding out whole sources



KEY IDEA

If a model will be rolled out to new sites, stores, countries or devices, hold out entire sources. This is sometimes called **out-of-distribution** or **external** validation.

IN THE WILD

Imaging models trained at one hospital have learned scanner-specific markings that do not exist elsewhere. External validation at a different hospital is the only way to discover this before deployment.

SECTION 3 RECAP

Splitting your data

REMEMBER

Takeaways

- Train to learn, validate to choose, test once to confirm
- Stratify imbalanced labels
- Split by group when rows are related
- Split by time when predicting the future, with a gap
- Hold out whole sites for external validation

CHECK YOURSELF

You are predicting which customers will buy next month. Each customer has many monthly rows. How should you split?

ANSWER

By time first, so the test months come after training months; consider grouping by customer too if new customers matter.

SECTION 04

04 Cross-validation in depth

Reusing data efficiently to get stable estimates, and the many variants for grouped, imbalanced and time-ordered data.

K-fold · Stratified · Repeated · Leave-one-out · Group · Time series · Nested

THE IDEA

Why cross-validation exists

A single hold-out split gives one estimate from one particular slice of data. With small or medium datasets, that estimate can swing a lot depending on which rows happened to land in the test set.

KEY IDEA

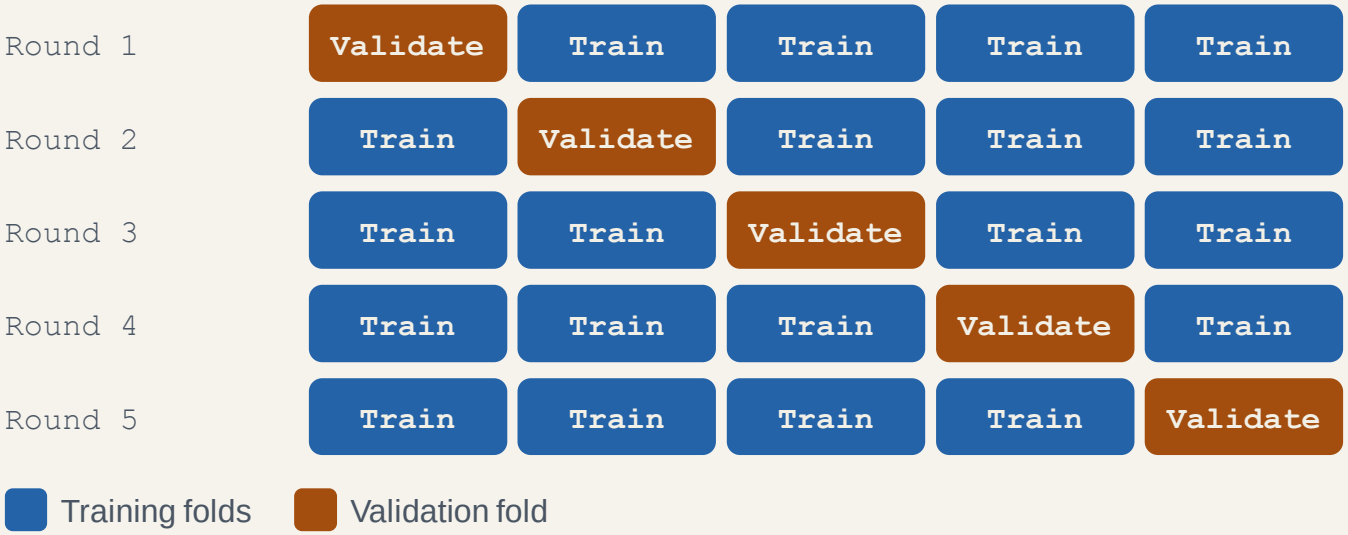
Cross-validation rotates the role of held-out data, so every row is used for checking exactly once. You get several estimates and can see how much they vary.

ANALOGY

Instead of judging a chef on one dish, you ask them to cook five different meals and look at the average and the spread. One bad night no longer decides the verdict.

THE WORKHORSE

K-fold cross-validation



1. **Shuffle and split**
   Divide data into K equal folds.

2. **Rotate**
   Train on K−1 folds, score the one left out.

3. **Repeat K times**
   Each fold is held out once.

4. **Summarise**
   Report the average and the spread.

THE WORKHORSE

Choosing K

K	Training data per round	Character	Good for
3	About two thirds	Fast, a little pessimistic	Big data, slow models
5	Four fifths	Balanced default	Most problems
10	Nine tenths	Less pessimistic, more compute	Small to medium data
N (one per row)	All but one row	Very slow, can be unstable	Tiny datasets only

RULE OF THUMB

Start with 5 or 10 folds. Larger K means each model trains on more data, so estimates are less pessimistic, but you pay with more training runs.

PITFALL

The fold scores are not independent: the training sets overlap heavily. So the spread across folds understates the true uncertainty somewhat.

VARIANT

Stratified k-fold

DEFINITION

Each fold keeps the same proportion of each class as the whole dataset.

USE IT WHEN

Any classification task, and especially when one class is rare.

PITFALL

With very rare classes, some folds may contain only a handful of positives. Check counts per fold, and reduce K if needed.

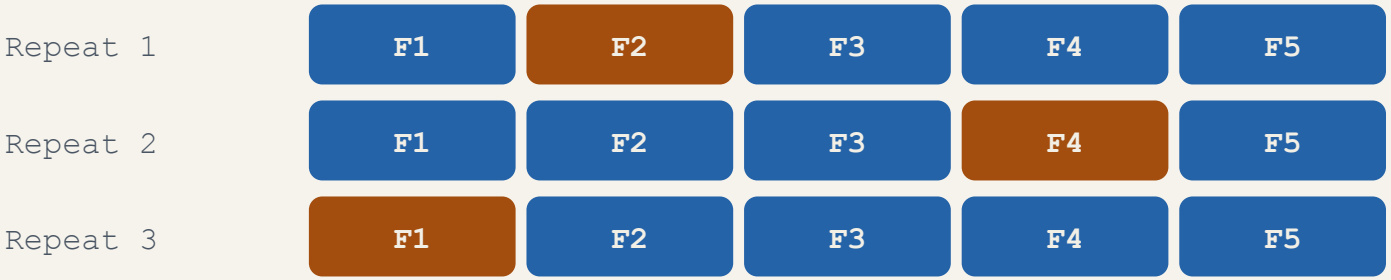
PYTHON · SCIKIT-LEARN

```
from sklearn.model_selection import StratifiedKFold, cross_val_score

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
scores = cross_val_score(model, X, y, cv=cv, scoring="average_precision")
print(scores.mean(), scores.std())
```

VARIANT

Repeated k-fold



Each repeat reshuffles the data before folding, so different rows end up together.

KEY IDEA

Running k-fold several times with different shuffles averages away the luck of one particular partition. Estimates become more stable.

USE IT WHEN

Small datasets, close comparisons between models, or results that move a lot with the random seed.

VARIANT

Leave-one-out and its cousins

Leave-one-out

Each row is the test set once. Uses almost all data for training, but needs as many fits as rows and can give jumpy estimates.

Leave-one-group-out

Each whole group (patient, store, site) is held out once. Ideal when groups are few and important.

Leave-p-out

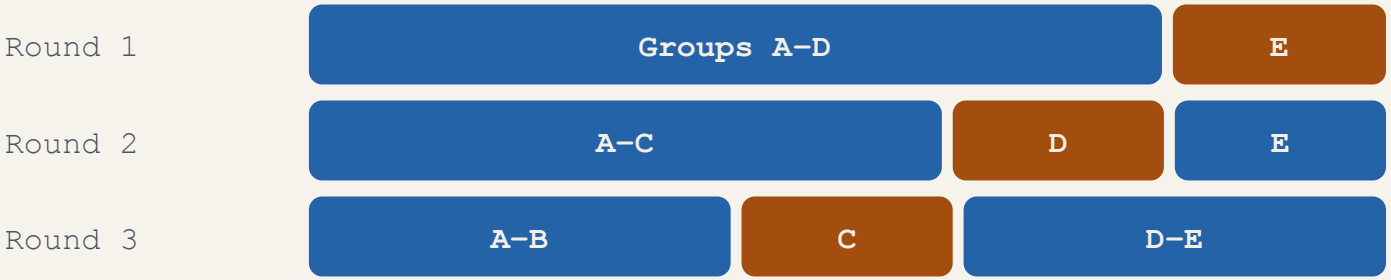
Every possible set of p rows is held out. Rarely practical beyond tiny data; mainly of theoretical interest.

PRO TIP

Leave-one-group-out is the go-to for multi-site studies: train on all hospitals but one, test on the excluded hospital, rotate. It directly measures transfer to an unseen site.

VARIANT

Group k-fold



Folds are made of whole groups, so no group is split between training and validation.

USE IT WHEN

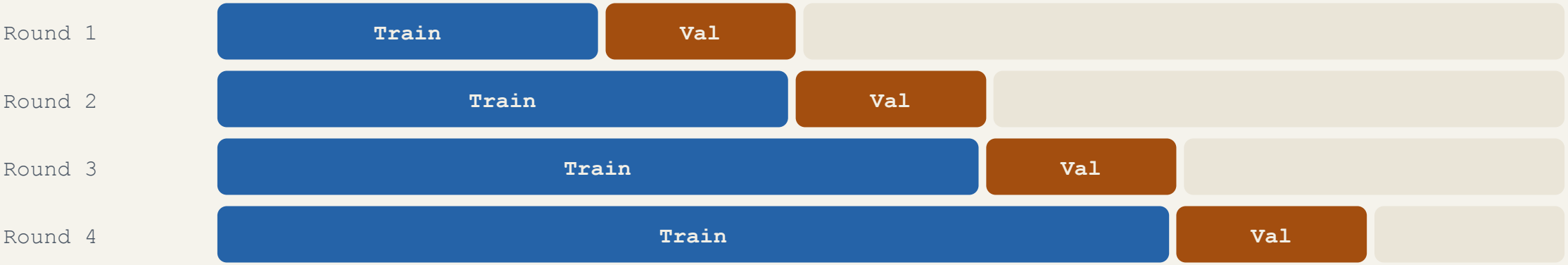
Rows from the same patient, user, device, session or document.

PRO TIP

Use **stratified group k-fold** when you need both: whole groups per fold and similar class balance across folds.

VARIANT

Time-series cross-validation



DEFINITION

Expanding window: training always starts at the beginning and grows; validation is always the next period.

DEFINITION

Sliding window: training is a fixed-length recent window that moves forward. Better when old data is no longer relevant.

KEY IDEA

Validation always comes after training. The model never sees the future it is asked to predict.

VARIANT

Purging and embargo for overlapping labels



DEFINITION

When each label spans a period (for example, “price rises over the next 10 days”), training rows near the validation window share information with it. **Purging** removes those overlapping rows; an **embargo** adds a buffer after validation.

USE IT WHEN

Finance, sensor data, any problem where outcomes are defined over windows that can overlap.

THE ADVANCED MUST-KNOW

Nested cross-validation

If you tune hyperparameters with cross-validation and then report the best cross-validated score, that score is optimistic: you picked the winner on the same folds you measured it with.

DEFINITION

Nested CV wraps tuning inside an outer loop. The inner loop chooses settings; the outer loop scores the whole tuning procedure on data it never touched.

1. **Outer split**
   Hold out one outer fold.
2. **Inner tuning**
   Run CV on the remaining folds to pick settings.
3. **Refit**
   Train with the chosen settings on all inner data.
4. **Score**
   Evaluate on the outer fold. Repeat for each.

THE ADVANCED MUST-KNOW

Nested cross-validation in code

PYTHON · SCIKIT-LEARN

```
from sklearn.model_selection import GridSearchCV, cross_val_score,
KFold

inner = KFold(n_splits=3, shuffle=True, random_state=1)
outer = KFold(n_splits=5, shuffle=True, random_state=2)

search = GridSearchCV(pipeline, param_grid, cv=inner,
                      scoring="roc_auc")

# each outer fold re-runs the whole search
scores = cross_val_score(search, X, y, cv=outer,
                         scoring="roc_auc")
```

KEY IDEA

The outer score estimates how well **your whole procedure** (including tuning) performs, not one particular setting.

PRO TIP

Nested CV tells you what to expect. For the final model, re-run the inner search on all available training data.

CHOOSING A SCHEME

Which cross-validation should I use?

Your data looks like…	Use	Why
Independent rows, balanced labels	K-fold (5 or 10)	Simple and reliable
Independent rows, imbalanced labels	Stratified k-fold	Every fold has rare cases
Rows grouped by entity	Group or stratified group k-fold	No entity on both sides
Time-ordered	Expanding or sliding window	Never train on the future
Overlapping time labels	Purged CV with embargo	Removes boundary leakage
Small data and tuning	Nested, repeated CV	Honest estimate of the procedure
Few sites, deploy to new ones	Leave-one-group-out	Measures transfer to unseen sites

COMMON MISTAKES

Cross-validation traps

⚠ PITFALL

Scaling, imputing or selecting features on the full dataset before splitting into folds.

⚠ PITFALL

Oversampling minority cases before CV, so copies of the same row land in training and validation.

⚠ PITFALL

Reporting the best of many tuned configurations as if it were an unbiased estimate.

⚠ PITFALL

Shuffling time-series data into random folds.

⚠ PITFALL

Forgetting groups, so the same patient appears in different folds.

⚠ PITFALL

Reporting only the mean and hiding a large fold-to-fold spread.

🔧 THE FIX

Put every data-dependent step inside a **pipeline** that is fitted separately within each training fold.

PRACTICAL CODE

The pipeline pattern

PYTHON · SCIKIT-LEARN

```
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression

pipe = Pipeline([
    ("impute", SimpleImputer(strategy="median")),
    ("scale", StandardScaler()),
    ("model", LogisticRegression(max_iter=1000)),
])
# imputer and scaler are re-fitted inside every fold
cross_val_score(pipe, X, y, cv=cv)
```

KEY IDEA

A pipeline guarantees that preprocessing learns only from the training part of each fold.

PRO TIP

Treat anything with a fit step (encoders, imputers, selectors, samplers, text vectorisers) as part of the model.

REPORTING

How to report cross-validation results

DO

- Report the mean **and** the spread across folds
- State the scheme: type, K, repeats, grouping
- Show the per-fold scores in an appendix
- Mention whether tuning was nested

AVOID

- Quote one rounded number with no context
- Hide unstable folds
- Compare models evaluated on different folds

WORKED EXAMPLE

A good one-liner

“Mean ROC AUC 0.87 across 5×5 repeated stratified group k-fold (grouped by customer), fold range 0.84 to 0.90. Hyperparameters tuned with nested CV.”

PRO TIP

When comparing models, use **the same folds** for all of them, so differences reflect the models, not the splits.

SECTION 4 RECAP

Cross-validation in depth

REMEMBER

Takeaways

- Rotate held-out data to get stable estimates
- Match the scheme to the data: stratified, group, time
- Nest tuning inside an outer loop for honest estimates
- Put every fitted step inside a pipeline
- Report spread, not just the mean

CHECK YOURSELF

You tune 200 configurations with 5-fold CV and report the best mean score. Why is that score optimistic?

ANSWER

You selected the configuration that did best on those exact folds, partly by luck. Nested CV or a fresh test set gives the honest number.

SECTION 05

05 Classification metrics

Every major way to score a classifier, what each one rewards, and when each one misleads.

- Confusion matrix
- Precision and recall
- F-scores
- ROC and PR curves
- Log loss
- MCC
- Multi-class

BEFORE ANY FORMULA

Choosing a metric is choosing which mistakes you care about.

There is no neutral metric. Each one weighs false alarms and missed cases differently. Pick the one whose weighting matches the real-world cost.

THE FOUNDATION

The confusion matrix

	Predicted: positive	Predicted: negative
Actual: positive	**True positive** Correctly flagged a real case	**False negative** Missed a real case
Actual: negative	**False positive** Raised a false alarm	**True negative** Correctly left alone

KEY IDEA

Every classification metric is a different summary of these four counts.

PRO TIP

Always look at the raw matrix before any single score. It shows you exactly where the errors are.

TWO ERROR TYPES

False positives versus false negatives

Application	A false positive means	A false negative means	Usually worse
Cancer screening	Unneeded follow-up test	Missed cancer	False negative
Spam filter	Real email hidden	Spam in inbox	False positive
Fraud detection	Blocked honest customer	Fraud gets through	It depends on amounts
Content moderation	Legitimate post removed	Harmful post stays up	Context dependent
Predictive maintenance	Unneeded inspection	Unexpected breakdown	False negative

ASK YOURSELF

Before choosing any metric, write down: what does each error cost, and who bears that cost?

THE TEMPTING ONE

Accuracy

DEFINITION

The share of all predictions that were correct.

USE IT WHEN

Classes are reasonably balanced and both error types cost about the same.

AVOID IT WHEN

Classes are imbalanced. With 1% positives, predicting “negative” for everyone scores 99%.

IN THE WILD

A fraud model reported 99.8% accuracy. Fraud was about 0.2% of transactions. The model had learned to say “not fraud” every time. Accuracy was the wrong lens entirely.

QUALITY OF ALARMS

Precision

DEFINITION

Of all the cases the model flagged as positive, the share that really were positive.

IN PLAIN WORDS

“When the model raises the alarm, how often is it right?”

USE IT WHEN

False alarms are costly: manual review queues, customer friction, wasted interventions.

ANALOGY

A smoke alarm with high precision rarely goes off for burnt toast. When it sounds, you believe it.

PITFALL

A model can get very high precision by flagging only the few most obvious cases and missing everything else.

COVERAGE OF CASES

Recall (sensitivity)

DEFINITION

Of all the cases that really were positive, the share the model caught.

IN PLAIN WORDS

“Of everything we should have caught, how much did we catch?”

USE IT WHEN

Missing a case is costly: disease, safety faults, serious fraud.

ANALOGY

A fishing net with high recall catches nearly every fish in the area, along with plenty of weeds.

PITFALL

Flagging everything gives perfect recall. Recall must always be read alongside precision or false-alarm rate.

THE OTHER SIDE

Specificity and the false-alarm rate

DEFINITION

Specificity: of all truly negative cases, the share the model correctly left alone.

DEFINITION

False-positive rate: of all truly negative cases, the share wrongly flagged. It is the flip side of specificity.

DEFINITION

Negative predictive value: when the model says “no”, how often it is right.

KEY IDEA

Medicine traditionally pairs **sensitivity and specificity**. Information retrieval and most ML teams pair **precision and recall**. They answer different questions; both are valid.

THE TRADE-OFF

Precision and recall pull against each other

Most classifiers output a score. A threshold turns the score into a yes or no. Lower the threshold and you catch more cases (recall up) but raise more false alarms (precision down).

Threshold	Flags	Precision	Recall
Very low	Almost everything	Low	Very high
Medium	Moderate number	Medium	Medium
Very high	Only the surest cases	Very high	Low

RULE OF THUMB

You do not have “a precision” or “a recall”. You have a precision and recall **at a chosen threshold**. Always state it.

PRO TIP

Ask the business for a constraint, such as “the review team can handle 200 cases a day”. That fixes the threshold, and you then maximise recall within it.

COMBINING TWO NUMBERS

F1 and the F-beta family

DEFINITION

F1 combines precision and recall into one score that is only high when both are high. A weakness in either pulls it down sharply.

DEFINITION

F-beta lets you lean one way: settings that favour recall suit screening; settings that favour precision suit costly interventions.

USE IT WHEN

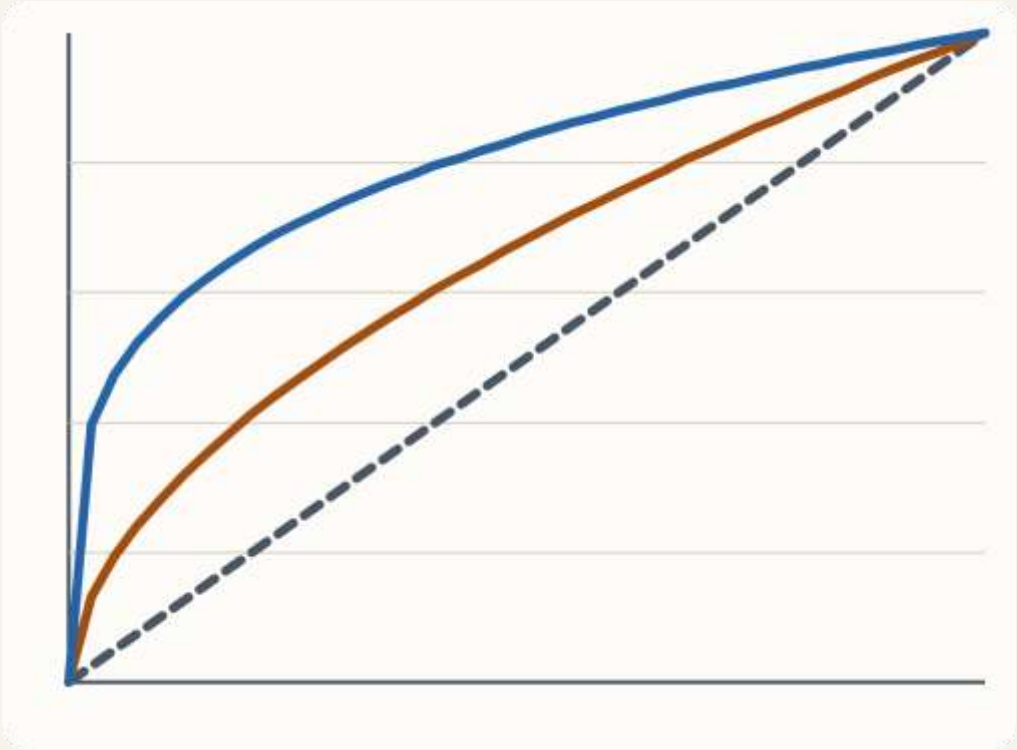
You need one number to rank models and both error types matter.

PITFALL

F1 ignores true negatives completely, depends on the threshold, and treats both errors as equally costly. That last assumption is rarely true.

THRESHOLD-FREE VIEW

The ROC curve and ROC AUC



Across: false-alarm rate · Up: recall

Strong model

Weaker model

Random guessing

DEFINITION

The ROC curve traces recall against false-alarm rate across every threshold. **ROC AUC** is the area under it.

IN PLAIN WORDS

ROC AUC is the chance that a randomly chosen positive gets a higher score than a randomly chosen negative.

THRESHOLD-FREE VIEW

When ROC AUC misleads

KEY IDEA

ROC AUC measures **ranking** quality only. It is unaffected by class balance and says nothing about whether scores are well calibrated.

PITFALL

With rare positives, a model can have a high ROC AUC yet produce mostly false alarms at any useful threshold, because a tiny false-alarm rate on millions of negatives is still a flood.

USE IT WHEN

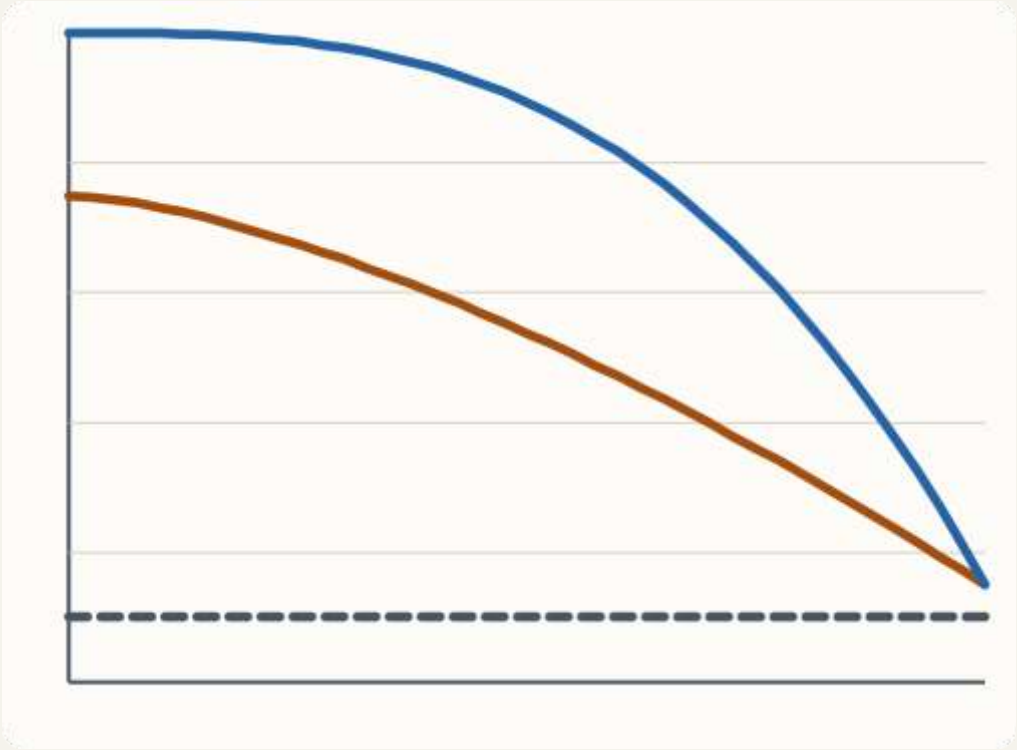
Classes are reasonably balanced, or you care about ranking across the whole range.

RULE OF THUMB

If positives are rare and you care about the flagged cases, prefer the precision–recall curve.

THRESHOLD-FREE VIEW

Precision–recall curve and average precision



Across: recall · Up: precision

Strong model

Weaker model

Baseline = positive rate

DEFINITION

The PR curve shows precision at every level of recall. **Average precision** (PR AUC) summarises it.

KEY IDEA

Its baseline is the positive rate, not one half. With 1% positives, 0.30 can be a strong result.

SCORING PROBABILITIES

Log loss and the Brier score

DEFINITION

Log loss punishes confident wrong answers very heavily. Saying 99% sure and being wrong costs far more than saying 60% sure and being wrong.

DEFINITION

Brier score is the average squared gap between predicted probability and what happened. Gentler on confident mistakes; easy to explain.

USE IT WHEN

Probabilities themselves are used: pricing, risk scores, expected-value decisions, combining models.

PRO TIP

These are **proper scoring rules**: a model scores best by reporting its honest belief. That makes them ideal for training and comparing probabilistic models.

BALANCED SUMMARIES

Matthews correlation and Cohen’s kappa

DEFINITION

MCC uses all four cells of the confusion matrix. It is high only when the model does well on both classes, so it resists being fooled by imbalance.

DEFINITION

Cohen’s kappa measures agreement beyond what chance alone would produce. Often used to compare models with human annotators.

WORKED EXAMPLE

A model that always predicts the majority class can have accuracy of 99% but an MCC of zero. MCC immediately reveals that it has learned nothing.

RULE OF THUMB

If you must report a single threshold-dependent number for an imbalanced binary problem, MCC is a robust choice.

MORE THAN TWO CLASSES

Averaging across classes

Averaging	How it works	Effect	Use when
Macro	Score each class, then simple average	Every class counts equally	Rare classes matter
Weighted	Average weighted by class size	Big classes dominate	Overall picture matters
Micro	Pool all decisions, then score	Equals accuracy for single-label	Every decision counts equally
Per class	Report each class separately	Nothing hidden	Always, in an appendix

⚠ PITFALL

Weighted and micro averages can look excellent while a small but important class is almost never predicted correctly. Always inspect the per-class table.

MORE THAN TWO CLASSES

Top-k accuracy and multi-label tasks

DEFINITION

Top-k accuracy: the right answer appears among the model's k highest-ranked guesses. Useful when a human chooses from a shortlist.

DEFINITION

Multi-label tasks allow several correct labels per example. Use per-label metrics, subset accuracy (all labels exactly right), or Hamming loss (share of label decisions wrong).

IN THE WILD

A support-ticket router suggests three possible teams to an agent. Top-1 accuracy is modest, but top-3 accuracy is high, and that matches how the tool is actually used.

METRIC CHOOSER

Which classification metric?

Situation	Primary metric	Also report
Balanced, equal costs	Accuracy or F1	Confusion matrix
Rare positives, alerts reviewed	Average precision; precision at fixed workload	Recall at that threshold
Missing cases is dangerous	Recall at a minimum precision	False-alarm rate
Ranking quality overall	ROC AUC	PR curve if imbalanced
Probabilities drive decisions	Log loss or Brier score	Calibration plot
Single robust number, imbalanced	MCC	Per-class results
Many classes, rare ones matter	Macro F1	Per-class table

SECTION 5 RECAP

Classification metrics

REMEMBER

Takeaways

- Start from the confusion matrix and real error costs
- Accuracy fails under imbalance
- Precision and recall depend on the threshold
- ROC AUC ranks; PR AUC focuses on positives
- Proper scoring rules judge probabilities

CHECK YOURSELF

Positives are 0.5% of data. Model A has ROC AUC 0.95 and average precision 0.12. Model B has ROC AUC 0.93 and average precision 0.31. Which is better for an alert queue?

ANSWER

Usually B. Its flagged cases are far more often real, which is what an alert queue cares about.

SECTION 06

06 Regression metrics

Measuring numeric predictions: error sizes, percentage errors, explained variation and the residual plots that reveal what numbers hide.

MAE | RMSE | Percentage errors | R-squared | Quantile loss | Residuals

THE BASICS

Average size of the error

DEFINITION

MAE

MAE (mean absolute error): the typical size of a miss, in the same units as the target. Every error counts in proportion to its size.

DEFINITION

RMSE

RMSE (root mean squared error): also in target units, but large errors count disproportionately. One huge miss inflates it a lot.

USE IT WHEN

MAE: when a miss of 20 is simply twice as bad as a miss of 10, and outliers should not dominate.

USE IT WHEN

RMSE: when big misses are especially harmful, such as under-stocking critical supplies.

PRO TIP

Report both. If RMSE is much larger than MAE, a few large errors are driving the result; go and find them.

ROBUST CHOICES

Median error and Huber-style losses

DEFINITION

Median absolute error: the middle miss. Almost unaffected by a handful of extreme errors.

DEFINITION

Huber-style losses: behave like squared error for small misses and like absolute error for big ones. A compromise between MAE and RMSE.

IN THE WILD

House-price predictions include a few mansions worth many times the typical home. RMSE is dominated by them. Median error describes the typical buyer’s experience, while a separate high-value slice reports the mansions.

RELATIVE ERRORS

Percentage errors and their traps

DEFINITION

MAPE: the average error as a percentage of the true value. Easy to explain to business users.

PITFALL

Explodes when true values are near zero; punishes over-forecasts and under-forecasts unequally; meaningless for targets that can be negative.

THE FIX

Symmetric MAPE reduces the asymmetry. **Weighted percentage error** (total absolute error divided by total actual) avoids division by tiny values and suits sales and demand data.

RULE OF THUMB

Never use MAPE on intermittent demand with many zero days. Use a scaled error instead.

EXPLAINED VARIATION

R-squared, carefully

DEFINITION

R-squared: how much of the variation in the target the model explains, compared with always predicting the average. One is perfect; zero matches the average; it can go negative on new data.

IN PLAIN WORDS

“How much better than just guessing the mean are we?”

PITFALL

A high R-squared does not mean predictions are accurate enough to be useful, and a low one does not mean the model is useless. It depends on the scale of the errors that matter.

PITFALL

R-squared on training data always rises as you add features. Only out-of-sample values mean anything.

SCALE-FREE COMPARISON

Scaled errors for forecasting

DEFINITION

MASE compares the model's error with the error of a simple naive forecast on the same series. Below one beats the naive method; above one loses to it.

WHY IT MATTERS

It works across series of very different sizes and never divides by zero, so you can average it over thousands of products.

WORKED EXAMPLE

A retailer forecasts 5,000 products. Raw errors are dominated by the best-sellers. A scaled error tells you, product by product, whether the model beats "same as last season".

PREDICTING RANGES

Quantile (pinball) loss and interval coverage

DEFINITION

Quantile loss scores a prediction like “90% of the time demand will be below this”. It penalises under-predictions and over-predictions with different weights.

DEFINITION

Interval coverage checks whether, say, 80% prediction intervals actually contain the truth about 80% of the time.

KEY IDEA

When decisions depend on risk (stock levels, staffing, capacity), validate the **range** a model predicts, not only its central estimate.

PITFALL

Intervals that are far too wide always cover the truth. Check sharpness (how narrow they are) as well as coverage.

THE MOST IMPORTANT PLOT

Residual analysis

Residuals vs prediction

A funnel shape means errors grow with the size of the prediction. Consider modelling on a log scale.

Residuals vs each feature

A curve or pattern means the model is missing structure in that feature.

Residuals over time

A trend or jump suggests drift or a change in the process.

Distribution of residuals

Heavy tails or skew show where averages hide extreme misses.

KEY IDEA

Metrics summarise. Residual plots diagnose. A good regression validation always includes at least the first and third plots.

BUSINESS FRAMING

Asymmetric costs in regression

IN THE WILD

Hospital capacity

Under-forecasting hospital bed demand means patients wait; over-forecasting means idle staff. The costs are unequal, so a symmetric metric like RMSE misrepresents what the hospital cares about.

THE FIX

Use a cost-weighted error, or quantile loss at the level that matches the cost ratio. Report over- and under-prediction separately.

PRO TIP

Ask stakeholders: “Would you rather be over by 10 or under by 10?” If they hesitate, errors are asymmetric and your metric should be too.

METRIC CHOOSER

Which regression metric?

Situation	Primary metric	Also report
Typical error matters, outliers are noise	MAE or median error	Residual distribution
Large misses are very costly	RMSE	Worst-case errors
Stakeholders think in percentages	Weighted percentage error	MAE by segment
Many series of different scales	MASE or scaled error	Per-series breakdown
Risk-based decisions	Quantile loss, interval coverage	Interval width
Unequal over/under costs	Cost-weighted error	Bias (average signed error)

SECTION 6 RECAP

Regression metrics

REMEMBER

Takeaways

- MAE for typical error; RMSE when big misses matter
- Percentage errors break near zero
- R-squared is relative, not absolute
- Validate prediction ranges, not just points
- Always inspect residual plots

CHECK YOURSELF

RMSE is three times larger than MAE on your test set. What does that suggest?

ANSWER

A small number of very large errors. Find them: they may be outliers, data errors or a segment the model handles badly.

SECTION 07

07 Metrics for other tasks

Validation beyond plain classification and regression: ranking, recommendation, clustering, vision, language, forecasting and anomaly detection.

Ranking · Recommenders · Clustering · Vision · NLP · Forecasting · Anomalies

RANKING AND SEARCH

Order matters

DEFINITION

Precision at k: of the top k results, how many are relevant.

DEFINITION

Recall at k: of all relevant items, how many appear in the top k.

DEFINITION

MRR: how high, on average, the first relevant result appears.

DEFINITION

NDCG: rewards placing the most relevant items nearest the top, with credit fading further down.

DEFINITION

MAP: average precision across many queries, rewarding relevant items ranked early.

DEFINITION

Hit rate: share of users or queries with at least one relevant item in the top k.

RULE OF THUMB

Choose k to match the screen: if users see five results, evaluate at five.

RECOMMENDERS

Beyond accuracy in recommendation

A recommender can score well on offline ranking metrics and still produce a dull, repetitive or unfair experience.

> ⚠ **PITFALL**
>
> Offline data only contains feedback on items the old system chose to show. A new model is judged on a biased record of the past.

Quality	Question
Coverage	How much of the catalogue ever gets recommended?
Diversity	Are the suggestions varied within a list?
Novelty	Does it surface less obvious items?
Serendipity	Pleasant surprises users would not find alone?
Popularity bias	Is it just recommending best-sellers?

Offline evaluation of logged decisions

DEFINITION

Off-policy evaluation estimates how a new policy would have performed using logs from the old one, by re-weighting outcomes according to how likely each choice was under each policy.

PITFALL

These estimates can be very noisy when the new policy chooses actions the old one rarely took. They also need the old system’s choice probabilities to have been logged.

PRO TIP

Log the probability with which every recommendation was shown, and keep a small amount of randomised exploration in production. Both make future offline validation far more reliable.

CLUSTERING

Validating without labels

DEFINITION

Internal measures (silhouette and similar) check whether clusters are tight and well separated, using only the data itself.

DEFINITION

External measures (adjusted Rand index, mutual information) compare clusters with known labels, when you have some.

KEY IDEA

The real test is **usefulness and stability**: do clusters stay similar when you resample the data, and do domain experts find them meaningful and actionable?

PITFALL

Internal scores favour certain shapes of cluster. A high silhouette score on the wrong representation is still the wrong answer.

COMPUTER VISION

Detection and segmentation

IoU

How much a predicted box or mask overlaps the true one, relative to their combined area.

mAP

Average precision across classes, often across several overlap thresholds. The standard for object detection.

Dice score

Overlap measure common in medical segmentation, focusing on the region of interest.

Small objects

Report performance by object size; small items are often where models fail.

PITFALL

A detection model’s mAP depends heavily on the overlap threshold chosen. Always state it, and compare models at the same setting.

LANGUAGE TASKS

Classic NLP metrics

Task	Common metrics	Watch out for
Text classification	F1, macro F1, MCC	Label imbalance, annotator disagreement
Named entities	Entity-level precision, recall, F1	Partial matches counted inconsistently
Translation	BLEU, chrF, COMET-style learned metrics	Surface overlap ignores meaning
Summarisation	ROUGE, learned metrics, human ratings	Fluent but unfaithful summaries
Question answering	Exact match, token F1	Correct answers phrased differently

KEY IDEA

Overlap metrics reward matching words, not being right. For generative output, pair them with human or model-based judgement, covered in section 15.

FORECASTING

Validating forecasts properly

KEY IDEA

Use **rolling-origin evaluation**: repeatedly forecast from successive cut-off points and score each forecast against what actually happened next.

RULE OF THUMB

Evaluate at the forecast horizons that matter. A model great at next-day forecasts may be poor four weeks ahead.

- Compare against naive and seasonal naive
- Score each horizon separately
- Check promotions, holidays and shocks
- Validate intervals, not just points
- Use scaled errors across series

ANOMALY DETECTION

Validating rare-event detectors

⚠ PITFALL

True anomalies are rare and often unlabelled. The few labelled ones may not resemble the next attack or failure.

THE FIX

Combine sources of evidence: precision at the top of the ranked list, injected synthetic anomalies, expert review of flagged cases, and back-testing on known past incidents.

KEY IDEA

Report the alert volume at the chosen threshold. A detector that flags 5,000 events a day is unusable, whatever its recall.

IN THE WILD

A machine-monitoring model was tuned on past breakdowns. Experts reviewing its top 50 alerts each week found faults no label had recorded. Those reviews became the most valuable validation data.

SECTION 7 RECAP

Metrics for other tasks

REMEMBER

Takeaways

- Ranking: evaluate the top of the list users actually see
- Recommenders: also check diversity, coverage and logged bias
- Clustering: stability and usefulness beat internal scores
- Generative text: overlap metrics are not correctness
- Rare events: report alert volume and expert precision

CHECK YOURSELF

Your search results page shows ten links. Which metric family fits best?

ANSWER

Ranking metrics at ten: precision at 10, NDCG at 10, or MRR if the first good result matters most.

SECTION 08

08

Imbalance, thresholds and calibration

Turning scores into decisions you can trust: handling rare classes, choosing thresholds and checking that probabilities mean what they say.

Imbalanced data | Resampling | Threshold tuning | Cost curves | Calibration | Reliability plots

RARE OUTCOMES

What imbalance does to validation

⚠ PITFALL

Accuracy becomes almost meaningless.

⚠ PITFALL

Test sets may hold too few positives to estimate recall reliably.

⚠ PITFALL

Random folds can contain zero positives.

⚠ PITFALL

Small changes in a few cases swing metrics wildly.

⚠ PITFALL

Default thresholds of one half are almost always wrong.

⚠ PITFALL

Resampling done carelessly leaks into validation.

RULE OF THUMB

Count your positives, not your rows. A test set with 40 positive cases cannot pin down recall to within a few percentage points, however many negatives it has.

RARE OUTCOMES

Resampling without fooling yourself

DEFINITION

Oversampling repeats or synthesises minority cases; **undersampling** drops majority cases; **class weights** make rare mistakes cost more during training.

KEY IDEA

Resampling changes training. It must **never** be applied to validation or test data, which must keep the real-world class balance.

PYTHON · IMBALANCED-LEARN

```
from imblearn.pipeline import Pipeline
from imblearn.over_sampling import SMOTE

pipe = Pipeline([
    ("smote", SMOTE(random_state=0)),
    ("model", clf),
])
# SMOTE runs only on each training fold
cross_val_score(pipe, X, y, cv=cv,
    scoring="average_precision")
```

FROM SCORES TO DECISIONS

Threshold selection is part of the model

1. **Agree the objective**
   Maximise expected value, hit a minimum recall, or fit a review capacity.
2. **Tune on validation data**
   Sweep thresholds and compute the objective at each one.
3. **Freeze it**
   Record the chosen threshold with the model version.
4. **Confirm on the test set**
   Report all metrics at the frozen threshold.
5. **Monitor it live**
   Base rates drift; the best threshold drifts with them.

⚠ PITFALL

Choosing the threshold on the test set inflates every threshold-based metric you then report from it.

FROM SCORES TO DECISIONS

Cost-based thresholds

WORKED EXAMPLE

Fraud review

Each missed fraud costs about £400. Each false alarm costs about £8 in review time and customer friction. A false alarm is roughly fifty times cheaper than a miss, so the best threshold flags many more cases than a default would.

KEY IDEA

Put a cost on each cell of the confusion matrix, then choose the threshold that minimises total expected cost on validation data. This aligns the model directly with the business.

PRO TIP

Show stakeholders a simple table: at each candidate threshold, how many alerts per day, how many real cases caught, and the expected net saving. Decisions become concrete.

PROBABILITIES YOU CAN TRUST

What calibration means

DEFINITION

A model is **calibrated** when its probabilities match reality: of all cases it scores at 70%, about 70% turn out positive.

KEY IDEA

Good ranking and good calibration are different properties. A model can rank perfectly yet claim 99% when the truth is 60%.

ANALOGY

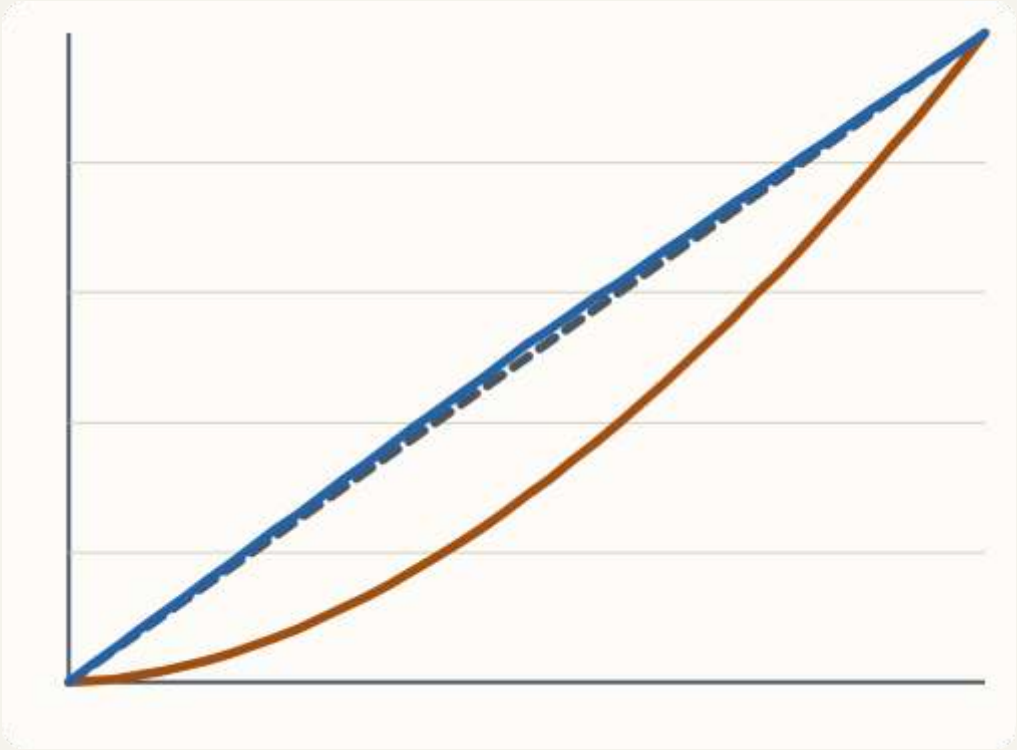
A well-calibrated weather forecaster says 30% chance of rain on many days, and it rains on about three in ten of them.

USE IT WHEN

Calibration matters when probabilities feed decisions: pricing, triage, expected-value maths, or being shown to users.

PROBABILITIES YOU CAN TRUST

Reliability diagrams



Across: predicted probability · Up: observed rate

Well calibrated

Overconfident

Perfect calibration

DEFINITION

Group predictions into bins by predicted probability and plot each bin’s average prediction against the share of real positives.

KEY IDEA

Points below the diagonal: the model is overconfident. Above it: underconfident.

PROBABILITIES YOU CAN TRUST

Measuring and fixing calibration

DEFINITION

Expected calibration error summarises the average gap between predicted and observed rates across bins. Simple, but sensitive to how bins are chosen.

THE FIX

Platt scaling fits a simple curve on top of scores; **isotonic regression** fits a flexible stepwise curve; **temperature scaling** is common for neural networks.

PITFALL

Fit calibration on separate data (a calibration split or cross-validation). Calibrating on the training data simply re-learns the original overconfidence.

PRO TIP

Recheck calibration after any resampling or class weighting: those techniques deliberately distort probabilities.

PROBABILITIES YOU CAN TRUST

Calibration under changing base rates

IN THE WILD

A model was calibrated when 2% of applications were fraudulent. During a fraud wave the rate tripled. Its 10% scores now corresponded to much higher real risk, and the fixed threshold let more fraud through.

THE FIX

Monitor predicted versus observed rates over time, and recalibrate on recent data when they diverge. Thresholds tied to costs may need to move too.

KEY IDEA

Calibration is a property of a model **and** a population. When the population changes, the calibration must be rechecked.

SECTION 8 RECAP

Imbalance, thresholds and calibration

REMEMBER

Takeaways

- Count positives; small counts mean wide uncertainty
- Resample training folds only, never test data
- Choose and freeze thresholds on validation data
- Link thresholds to real costs or capacity
- Check calibration with reliability diagrams

CHECK YOURSELF

You oversampled positives to a 50/50 balance before training. What happens to the model’s raw probabilities on real data?

ANSWER

They will be too high. Recalibrate on data with the real class balance before using them as probabilities.

SECTION 09

09 Data leakage

The silent killer of model validity: how information from the future or from the answer sneaks into training, and how to stop it.

- Target leakage
- Train-test contamination
- Temporal leakage
- Preprocessing leaks
- Detection
- Prevention

THE GOLDEN RULE

If a result looks too good to be true, assume leakage until proven otherwise.

Leakage gives models information they will not have at prediction time. It produces superb offline scores and disappointing live performance.

DEFINITIONS

Two families of leakage

Target leakage

A feature contains information about the outcome that would not be available when the prediction is actually made. The model cheats by reading the answer.

Train–test contamination

Information from validation or test data influences training, through shared rows, shared preprocessing, or repeated peeking.

ASK YOURSELF

For every feature, ask: “At the exact moment the model makes this prediction in production, would this value already be known, and would it have this exact value?”

TARGET LEAKAGE

Features that know the answer

Prediction task	Leaky feature	Why it leaks
Will patient develop sepsis?	Antibiotics prescribed	Often given because sepsis was suspected
Will customer churn?	Account closure reason	Only filled in after they leave
Will loan default?	Collections contact count	Collections start after missed payments
Is this claim fraudulent?	Investigation outcome code	Produced by the fraud team after review
Will the part fail?	Repair ticket opened	Opened when failure is already happening

KEY IDEA

Many leaky features are **consequences** of the outcome, not causes. Map the timeline of when each field is created.

TEMPORAL LEAKAGE

Information from the future

⚠ PITFALL

Random splits of time-ordered data.

⚠ PITFALL

Rolling averages that include the day being predicted.

⚠ PITFALL

Features computed from the whole dataset, such as an average over all years.

⚠ PITFALL

Data that was later corrected or backfilled used as if known at the time.

⚠ PITFALL

Labels defined over windows that overlap the training cut-off.

⚠ PITFALL

Using today's customer attributes for a prediction made last year.

THE FIX

Build features **as of** each prediction timestamp, using only data recorded before that moment. Feature stores with point-in-time joins exist to do exactly this.

PREPROCESSING LEAKAGE

Fitting on everything before splitting

PITFALL

Scaling, normalising or imputing using statistics from the full dataset lets test data shape the training inputs.

PITFALL

Selecting the “best” features using all labels, then cross-validating, gives famously inflated scores, especially with many features and few rows.

PITFALL

Target encoding categories using all rows puts the answer into the feature.

THE FIX

Split first. Fit every transformer on training data only, then apply it to validation and test data. Use pipelines so this happens automatically inside each fold.

WORKED EXAMPLE

With thousands of random, meaningless features and a few hundred rows, selecting features on all data before CV can produce impressive-looking accuracy on pure noise.

CONTAMINATION

Duplicates, near-duplicates and shared sources

⚠ PITFALL

The same record appears twice, once in each split. Common after data merges or augmentation.

⚠ PITFALL

Near-duplicates: the same photo resized, the same text lightly edited, consecutive video frames.

⚠ PITFALL

Augmented copies of one image placed in different splits.

THE FIX

Deduplicate **before** splitting, using exact hashes plus similarity search for near-duplicates. Augment only after splitting, and only the training set.

IN THE WILD

Benchmark datasets in vision and language have repeatedly been found to contain test items that also appear in training sets, sometimes with small edits. Published scores were inflated as a result.

CONTAMINATION

Leakage for large language models

KEY IDEA

Pre-trained models have seen vast amounts of web text. Public benchmark questions, and their answers, may have been in that text.

PITFALL

A strong benchmark score may reflect memorisation rather than ability, and you cannot fully inspect the training data.

THE FIX

Build private, freshly written evaluation sets; include items created after the model's training cut-off; paraphrase or perturb items and check if scores hold.

PRO TIP

If performance collapses when questions are lightly reworded, suspect memorisation.

LEAKAGE VIA PEOPLE

Human leakage: overfitting to the test set

Even with perfect pipelines, a team that runs hundreds of experiments and keeps whatever improves the test score will slowly fit the test set by hand.

ANALOGY

Like a student who retakes the same practice exam until they pass. The final score reflects familiarity with that paper, not mastery.

THE FIX

- Use validation data for day-to-day decisions
- Ration test-set evaluations and log them
- Refresh the test set periodically
- Keep a final “vault” set nobody sees until sign-off

DETECTION

How to spot leakage

Too-good scores

Performance far above baselines, published results or domain intuition.

One dominant feature

A single feature carries most of the model's importance.

Timeline review

For each top feature, when is it created relative to the prediction?

Adversarial validation

Train a model to tell training rows from test rows. If it succeeds easily, the sets differ in a suspicious way.

PRO TIP

Try training a model on each top feature alone. If one feature by itself nearly matches the full model, investigate it carefully before celebrating.

PREVENTION

A leakage-prevention checklist

- [ ] Write down the exact prediction moment for the use case
- [ ] Document when every feature becomes available
- [ ] Deduplicate before splitting, including near-duplicates
- [ ] Split by group and time where relevant
- [ ] Fit all preprocessing inside pipelines, per fold
- [ ] Augment and resample only training data
- [ ] Build features with point-in-time joins
- [ ] Run adversarial validation on train versus test
- [ ] Inspect the top features for plausibility
- [ ] Ration and log test-set evaluations

SECTION 9 RECAP

Data leakage

REMEMBER

Takeaways

- Leakage gives the model information it won’t have live
- Check feature timing against the prediction moment
- Fit preprocessing per fold, inside pipelines
- Deduplicate before splitting
- Human peeking at the test set is leakage too

CHECK YOURSELF

A hospital readmission model’s best feature is “discharge summary length”. Should you worry?

ANSWER

Check timing. If the summary is written at or after discharge and the prediction must be made earlier, it leaks. Even if available, it may proxy for complexity in ways worth understanding.

SECTION 10

10

Tuning and model selection

Choosing hyperparameters and models without overfitting the validation process itself.

Search strategies · Selection bias · Early stopping · Simplicity · Ensembles · Final refit

SEARCH STRATEGIES

How to search for good settings

Method	How it works	Strength	Weakness
Grid search	Tries every combination on a grid	Simple, exhaustive	Wasteful in many dimensions
Random search	Samples combinations at random	Efficient when few settings matter	No learning from results
Bayesian optimisation	Models the score and picks promising settings	Sample-efficient	More complex, sequential
Successive halving	Gives more budget to the best performers	Fast for big searches	Can drop slow starters
Evolutionary	Mutates and recombines good settings	Handles odd search spaces	Many evaluations

PRO TIP

Random search usually beats grid search for the same budget, because it explores more distinct values of the settings that actually matter.

A SUBTLE DANGER

Overfitting the validation set

The more configurations you try, the more likely one looks good by chance on your particular validation data. The winning score is biased upwards, and the bias grows with the number of trials.

ANALOGY

Flip a thousand coins ten times each. One will probably land heads nine or ten times. It is not a magic coin.

THE FIX

- Use nested CV or a separate untouched test set
- Prefer bigger validation sets for big searches
- Repeat CV to reduce noise
- Don’t chase tiny improvements

RULE OF THUMB

Any gain smaller than the fold-to-fold spread is probably noise.

Early stopping as validation

DEFINITION

Stop training when validation performance stops improving, and keep the best checkpoint.

PITFALL

The epoch you pick is chosen using validation data, so the final validation score is optimistic. Confirm on the test set.

PRO TIP

Use a patience setting so you don’t stop on a single noisy dip, and log the full curve for review.

WORKED EXAMPLE

Validation loss falls for 30 epochs, wobbles until epoch 45, then climbs steadily. With patience of 10, training stops at about epoch 55 and restores the checkpoint from the lowest point.

CHOOSING BETWEEN MODELS

Prefer simpler when it is close

RULE OF THUMB

The one-standard-error rule, in words

If a simpler model performs within the uncertainty of the best one, choose the simpler model. It is usually easier to explain, cheaper to run and more robust to change.

KEY IDEA

Model selection is not just about the top score. Weigh latency, cost, maintainability, interpretability, fairness and robustness alongside accuracy.

Criterion	Model A (gradient boosting)	Model B (logistic regression)
Validation score	Slightly higher	Within uncertainty of A
Latency and cost	Higher	Very low
Explainability	Needs extra tooling	Direct from coefficients
Likely choice		B, unless the gain is material

Validating combined models

PITFALL

Stacking trains a second model on the first models' predictions. If those predictions were made on their own training data, the stacker learns from overconfident inputs.

THE FIX

Generate first-level predictions **out-of-fold**: each row's prediction comes from a model that did not train on it. Then validate the whole stack with an outer loop.

PRO TIP

Ensembles often improve average scores. Check that they do not harm calibration or slow inference beyond what the product can accept.

AFTER SELECTION

The final refit

1. **Freeze the recipe**
   Features, model type, hyperparameters, threshold and preprocessing.
2. **Confirm once**
   Score the frozen recipe on the untouched test set.
3. **Decide on refitting**
   Optionally retrain on training plus validation data, or all data, to use every example.
4. **Sanity-check**
   Make sure the refit model behaves like the validated one on a holdout or recent sample.
5. **Record everything**
   Data version, code version, seeds and the test result.

⚠ PITFALL

Refitting on all data leaves nothing to test the final artefact. Keep a small recent holdout, or rely on shadow deployment to confirm it.

SECTION 10 RECAP

Tuning and model selection

REMEMBER

Takeaways

- Random or Bayesian search beats naive grids
- More trials means more optimistic best scores
- Early stopping uses validation data too
- Prefer simpler models when results are close
- Stack with out-of-fold predictions

CHECK YOURSELF

After 500 tuning runs, the best model beats the second best by a tiny margin. What should you do?

ANSWER

Treat them as tied. Check uncertainty, then decide on simplicity, cost and robustness rather than the tiny score gap.

SECTION 11

11

Uncertainty and statistical rigour

Turning single scores into honest ranges, and telling real improvements apart from noise.

- Confidence intervals
- Bootstrap
- Comparing models
- Significance
- Multiple comparisons
- Sample size

UNCERTAINTY

A score without a range is a rumour.

Rerun your evaluation on a different sample and the number will change. Validation must tell you by how much.

WHY IT MATTERS

Every metric is an estimate

Your test set is one sample of the world. A different sample of the same size would give a somewhat different score. The smaller the test set, or the rarer the event, the bigger that wobble.



ANALOGY

Polling 100 voters gives a rough answer. Polling 10,000 gives a much tighter one. Test sets work the same way.

KEY IDEA

Report a **confidence interval**: a range likely to contain the true performance. Narrow ranges support confident decisions; wide ranges call for more data.

THE PRACTICAL TOOL

Bootstrap confidence intervals

1. **Resample**
   Draw rows from the test set with replacement, same size as the original.
2. **Score**
   Compute the metric on the resampled set.
3. **Repeat**
   Do this a thousand or more times.
4. **Read the range**
   The middle 95% of the scores forms the interval.

KEY IDEA

The bootstrap works for almost any metric, including awkward ones like F1 or NDCG, without special formulas.

PITFALL

Resample at the right level. If rows are grouped by patient, resample **patients**, not rows, or the interval will be too narrow.

THE PRACTICAL TOOL

Bootstrap in code

PYTHON

```
import numpy as np
from sklearn.metrics import average_precision_score

rng = np.random.default_rng(0)
stats = []
for _ in range(2000):
    idx = rng.integers(0, len(y_test), len(y_test))
    if y_test[idx].sum() == 0:
        continue
    stats.append(average_precision_score(
        y_test[idx], scores[idx]))
low, high = np.percentile(stats, [2.5, 97.5])
```

PRO TIP

Use the same resampled indices for two models to get an interval on their **difference**. That is what actually decides between them.

WORKED EXAMPLE

“Average precision 0.41 (95% bootstrap interval 0.36 to 0.46, 2,000 resamples of customers).”

COMPARING MODELS

Is model B really better than A?

KEY IDEA

Compare models on **the same examples** (paired comparison). Shared easy and hard cases cancel out, so real differences show up much more clearly.

DEFINITION

McNemar's test compares two classifiers by looking only at the cases where they disagree. **Paired bootstrap** gives an interval on the score difference.

Question	Approach
Two classifiers, one test set	McNemar's test or paired bootstrap of the difference
Two models across CV folds	Corrected paired tests that account for overlapping training sets
Many models across many datasets	Rank-based tests with post-hoc comparisons
Two forecasts	Tests that compare forecast error series over time

SIGNIFICANCE IN PRACTICE

Statistical versus practical significance

DEFINITION

Statistically significant: the difference is unlikely to be pure chance.

DEFINITION

Practically significant: the difference is big enough to matter for users or the business.

WORKED EXAMPLE

With millions of test rows, a tiny difference can be statistically significant yet worthless. With a few hundred rows, a large and important difference may not reach significance.

RULE OF THUMB

Agree in advance on the **smallest improvement worth shipping**. Then ask whether the interval on the difference clears it.

HIDDEN TRAPS

Multiple comparisons and forking paths

⚠ PITFALL

Test twenty slices, metrics or variants and, by chance alone, about one will look significantly different.

⚠ PITFALL

Choosing which analysis to report after seeing results (the “garden of forking paths”) inflates confidence even without formal tests.

THE FIX

- Pre-register the main metric and comparison
- Adjust for multiple tests when hunting across many slices
- Treat unexpected findings as hypotheses to confirm on new data
- Report how many things you tried

PLANNING

How much test data is enough?

Goal	Rough guide
Estimate accuracy within a few points	Several hundred to a thousand cases
Estimate recall for a rare class	Hundreds of positive cases, not total rows
Detect a small difference between models	Many thousands of paired cases
Reliable results for each subgroup	Enough cases in each group, not just overall

PRO TIP

Run a quick bootstrap on a pilot sample. If the interval is too wide for the decision at hand, you need more test data before you can conclude anything.

VARIABILITY FROM TRAINING

Seeds and training noise

KEY IDEA

Uncertainty does not only come from the test sample. Retraining with a different random seed, data order or initialisation can change results, especially for neural networks.

THE FIX

Train several seeds and report the spread. Compare models across seeds, not from a single lucky run.

IN THE WILD

A new architecture appeared to beat the baseline by a clear margin. Across five seeds each, the ranges overlapped almost entirely. The “improvement” was a single fortunate initialisation.

SECTION 11 RECAP

Uncertainty and statistical rigour

REMEMBER

Takeaways

- Always report an interval, not just a point
- Bootstrap at the level of independent units
- Compare models on paired data
- Separate statistical from practical significance
- Account for multiple comparisons and seeds

CHECK YOURSELF

Model B beats A by 0.5 points; the 95% interval on the difference runs from −0.4 to +1.4. What do you conclude?

ANSWER

The evidence does not show B is better. The interval includes zero; you need more data or should treat them as equivalent.

SECTION 12

12

Error analysis and robustness

Looking beneath the average: finding failing slices, stress-testing inputs and checking behaviour under change.

Error analysis | Slicing | Stress tests | Behavioural tests | Adversarial | Shift

ERROR ANALYSIS

Read your mistakes

1. **Sample errors**
   Pull 50–100 misclassified or high-error cases.

2. **Tag them**
   Note why each went wrong: label error, ambiguity, missing feature, rare case.

3. **Count the tags**
   Rank causes by frequency and impact.

4. **Act on the biggest**
   Fix data, add features or change the approach.

KEY IDEA

An afternoon reading errors often teaches more than a week of tuning. Patterns appear that no metric reveals.

PRO TIP

Include the highest-confidence errors. Confident mistakes often expose label errors or leakage.

SLICING

Averages hide failing groups

Slice	Share of data	Recall	Note
Overall	100%	High	Looks ready to ship
New customers	8%	Much lower	Little history to learn from
Mobile app users	35%	Similar to overall	Fine
Non-English text	4%	Very low	Hidden by the average
Night-time transactions	12%	Lower	Different behaviour patterns

KEY IDEA

Decide the important slices **before** testing: by user type, region, device, language, time, data source and protected characteristics. Report each with its sample size and interval.

SLICING

Finding slices you did not think of

DEFINITION

Automated slice discovery searches combinations of features to find groups where error is unusually high, such as “older devices in rural areas on weekends”.

PITFALL

With many candidate slices, some will look bad by chance. Confirm discovered slices on fresh data before acting.

PRO TIP

Cluster the errors themselves, for example by embedding misclassified texts or images. Groups of similar failures often point to a missing concept in the training data.

STRESS TESTING

Robustness to messy inputs

Missing values

Blank fields, nulls in new places, missing sensors.

Noise

Typos, blur, compression, background noise in audio.

Format changes

New units, date formats, encodings, schema tweaks.

Extremes

Values far outside the training range.

KEY IDEA

Deliberately corrupt validation inputs in realistic ways and measure how gracefully performance degrades. A sudden collapse is a risk to report.

BEHAVIOURAL TESTING

Unit tests for model behaviour

Test type	What it checks	Example
Minimum functionality	Simple cases the model must get right	“This was great” is positive
Invariance	Irrelevant changes don’t change the output	Changing a name doesn’t change sentiment
Directional	Relevant changes move output the right way	Adding “not” flips sentiment
Monotonicity	More of a risk factor never lowers risk	Higher debt never reduces default risk

PRO TIP

Keep these tests in your CI pipeline. They catch regressions that aggregate metrics miss, and they document expected behaviour for reviewers.

ADVERSARIAL ROBUSTNESS

Inputs designed to fool the model

DEFINITION

Adversarial examples are inputs changed in small, deliberate ways that cause wrong predictions, often imperceptible to humans.

USE IT WHEN

Fraud, spam, content moderation, security, or any setting where people have a motive to game the model.

PITFALL

Validating only on natural data overstates robustness in adversarial settings. Attackers adapt to whatever the model does.

THE FIX

Red-team the model; test known evasion tactics; monitor for shifts in attack patterns after launch.

DISTRIBUTION SHIFT

Testing under change

KEY IDEA

Build validation sets that represent **plausible futures**, not just the past: new regions, new devices, seasonal peaks, economic shocks.

IN THE WILD

A demand model validated on two calm years failed during a sudden shock, when buying patterns changed overnight. A stress test using a past disruption would have exposed its fragility.

DEFINITION

Adversarial validation trains a classifier to separate training data from new data. Features that make this easy are where the shift lives.

PRO TIP

Hold out the most recent period, the most unusual period, and a new segment. Report all three alongside the standard test.

SECTION 12 RECAP

Error analysis and robustness

REMEMBER

Takeaways

- Read and categorise real errors
- Pre-define important slices and report each
- Corrupt inputs to test graceful degradation
- Write behavioural tests for expected logic
- Validate against plausible future shifts

CHECK YOURSELF

Overall recall is strong, but one region contributes 3% of data and shows very low recall. What do you do?

ANSWER

Report it explicitly with its interval, investigate why, and decide with stakeholders whether to fix, guard or limit use in that region.

SECTION 13

13

Fairness and responsible validation

Checking that a model’s errors and benefits are distributed acceptably across groups of people.

Group metrics · Trade-offs · Proxies · Intersections · Process

WHY IT BELONGS HERE

Fairness is a validation question

KEY IDEA

A model can be accurate overall and still make far more mistakes for one group than another. That is a validation failure: the evidence did not cover everyone the model affects.

WHY IT MATTERS

Unequal error rates can cause real harm, damage trust, and breach equality and data-protection law in many jurisdictions.

IN THE WILD

A recruitment screening tool learned from historic hiring decisions. Validation on overall accuracy looked fine. Slice analysis showed it systematically down-ranked applicants from groups that had been under-hired in the past.

MEASURING IT

Common group fairness measures

Measure	Asks whether…	Typical use
Demographic parity	Groups receive positive decisions at similar rates	Allocation of opportunities
Equal opportunity	Groups have similar recall among true positives	Benefits that should reach qualified people
Equalised odds	Groups have similar recall and false-alarm rates	High-stakes screening
Predictive parity	A flag means the same probability across groups	Risk scores used by humans
Calibration by group	Probabilities are accurate within each group	Any probability shown or acted on

HARD TRUTHS

You cannot satisfy every definition at once

KEY IDEA

When groups have different underlying rates of the outcome, several fairness measures mathematically cannot all be equal at the same time. Choosing which matters is a values decision, not a technical one.

THE FIX

Agree the fairness criterion with domain experts, affected communities, legal and ethics advisers **before** validation. Document the reasoning.

PRO TIP

Show decision-makers a table of each candidate criterion and what satisfying it would cost elsewhere. Make the trade-off visible rather than buried in a metric choice.

HIDDEN ROUTES

Proxies and intersections

⚠ PITFALL

Removing a protected attribute does not remove bias. Postcode, school, name or shopping habits can act as **proxies** for it.

⚠ PITFALL

A model can look fair for each group separately yet fail for **intersections**, such as older women or young men in one region.

THE FIX

Keep protected attributes available for **evaluation** (with appropriate safeguards), even if they are not used as model inputs. You cannot measure what you cannot see.

RULE OF THUMB

Check intersections that have enough data to be meaningful, and flag those that are too small to assess.

PROCESS

A fairness validation workflow

1. **Identify groups**
   Who is affected, and which groups could be harmed?
2. **Choose criteria**
   Agree which measures matter and acceptable gaps.
3. **Measure**
   Report each measure by group, with intervals.
4. **Investigate**
   Find causes: data, labels, features, thresholds.
5. **Mitigate and document**
   Fix, guard or limit use; record the decision.

KEY IDEA

Fairness validation is never finished at launch. Monitor group metrics in production, because populations and behaviour change.

SECTION 13 RECAP

Fairness and responsible validation

REMEMBER

Takeaways

- Overall accuracy can hide unequal errors
- Different fairness measures can conflict
- Choose criteria with stakeholders and document why
- Watch for proxies and intersections
- Keep monitoring after launch

CHECK YOURSELF

You removed gender from the inputs. Is the model now guaranteed to be gender-fair?

ANSWER

No. Other features can act as proxies. You still need to measure outcomes by gender to know.

14

SECTION 14

Explainability as a validation tool

Using explanations to check that a model is right for the right reasons.

Global importance | Local explanations | Sanity checks | Shortcut learning

RIGHT FOR THE RIGHT REASONS

Why explanations help validation

KEY IDEA

Metrics tell you **whether** a model is right. Explanations help you see **why**, and whether the reasons would hold up in the real world.

IN THE WILD

Shortcut learning

An image classifier for a disease scored very well. Explanations showed it focused on hospital labels and rulers in the images, not the anatomy. It had learned a shortcut.

ANALOGY

A student who gets the right answer by copying the person next to them scores well today. Asking them to show their working reveals the problem before the real exam.

TOOLS

Global and local explanations

Technique	Scope	What it shows	Caution
Permutation importance	Global	Score drop when a feature is shuffled	Misleading with correlated features
SHAP-style values	Global and local	Each feature’s contribution to a prediction	Explains the model, not the world
Partial dependence	Global	Average effect of changing one feature	Hides interactions
Counterfactuals	Local	Smallest change that flips the decision	May suggest unrealistic changes
Saliency maps	Local, images	Which pixels drove the output	Can be noisy or unstable

PRO TIP

Compute importance on **validation** data, not training data. Training-data importance can reward features the model merely memorised.

USING THEM WELL

Validation checks with explanations

ASK YOURSELF

Do the most important features make sense to a domain expert?

ASK YOURSELF

Is any feature suspiciously dominant, hinting at leakage?

ASK YOURSELF

Do effects point in sensible directions, such as more risk factors meaning more risk?

ASK YOURSELF

Do explanations look similar across groups, or does the model rely on proxies for some?

ASK YOURSELF

Are local explanations stable for near-identical inputs?

ASK YOURSELF

Would the model's reasoning survive a change in context?

PITFALL

Explanations are approximations. A plausible explanation does not prove the model is correct; it is one more piece of evidence.

SECTION 14 RECAP

Explainability as a validation tool

REMEMBER

Takeaways

- Use explanations to test for right reasons
- Compute importance on held-out data
- Dominant odd features often mean leakage
- Ask domain experts to review drivers
- Treat explanations as evidence, not proof

CHECK YOURSELF

Your top feature is a database row ID. What does that tell you?

ANSWER

Almost certainly leakage or a data-ordering artefact. IDs should carry no real-world meaning; investigate before anything else.

SECTION 15

15

Evaluating LLMs and generative models

Validating systems whose outputs are open-ended text, code or images, where there is rarely one correct answer.

Why it's different | Eval sets | Human rating | LLM-as-judge | RAG evaluation | Safety

WHAT CHANGES

Why generative evaluation is different

Many right answers

A good summary can be worded in countless ways, so exact matching fails.

Fluent but wrong

Outputs can sound confident and be factually incorrect.

Non-deterministic

The same prompt can give different outputs; evaluate several samples.

Contamination

Public benchmarks may sit in the training data.

KEY IDEA

Generative evaluation combines **task-specific test sets**, **clear rubrics**, **human judgement** and **automated judges**, each validating the others.

BUILDING BLOCKS

A good evaluation set for an LLM application

- Real user queries, sampled and anonymised
- Edge cases and tricky phrasings
- Questions with no answer in the sources
- Adversarial and unsafe prompts
- Coverage of every major intent
- Reference answers or grading notes

RULE OF THUMB

Start with 50–100 carefully curated examples that represent real use. A small, high-quality set beats a big, generic benchmark for product decisions.

PRO TIP

Version the evaluation set like code. Add every production failure you find as a new test case.

DEFINING GOOD

Rubrics turn opinions into measurements

Criterion	What a high score looks like
Correctness	Facts are accurate and consistent with the sources
Faithfulness	Every claim is supported by the provided context
Completeness	Covers all parts of the question
Relevance	Stays on topic, no padding
Safety and policy	Refuses harmful requests; no private data leaked
Tone and format	Matches the brand voice and required structure

PRO TIP

Use simple scales (pass/fail or 1–3) with written examples of each level. Fine-grained 1–10 scales produce noisy, inconsistent ratings

GOLD STANDARD

Human evaluation done properly

✓ DO

- Write clear guidelines with examples
- Blind raters to which system produced each output
- Randomise order in side-by-side comparisons
- Use several raters per item for a sample
- Measure agreement between raters

⚠ AVOID

- Let the developers be the only judges
- Rate on vague criteria like “quality”
- Ignore disagreement between raters
- Use too few items to see real differences

KEY IDEA

If trained raters disagree often, your rubric is ambiguous. Fix the rubric before trusting any scores, human or automated.

SCALING UP

LLM-as-judge

DEFINITION

A strong language model grades outputs against a rubric, either scoring one answer or choosing the better of two.

PITFALL

- May prefer longer answers
- May favour the first option shown
- May prefer its own style of writing
- Can be fooled by confident wording

USE IT WHEN

Fast iteration, regression testing, and large evaluation sets where human rating would be too slow.

THE FIX

Validate the judge: compare its ratings with human ratings on a sample, swap answer order, and recheck when the judge model changes.

Evaluating retrieval-augmented generation

1

Retrieval

Did we fetch the right passages?

2

Generation

Did the answer use them faithfully?

3

End to end

Did the user get a correct, helpful answer?

DEFINITION

Context recall and precision: relevant passages found, and little irrelevant noise.

DEFINITION

Faithfulness or groundedness: claims supported by the retrieved context.

DEFINITION

Answer correctness: matches the reference or rubric.

PRO TIP

Evaluate retrieval and generation separately. When the final answer is wrong, you need to know which stage failed.

SAFETY

Red-teaming and safety evaluation

KEY IDEA

Generative systems must be validated for what they should **not** do: harmful content, leaking private data, following injected instructions, or making up citations.

DEFINITION

Red-teaming means people (and automated tools) deliberately trying to make the system misbehave, then turning each success into a regression test.

- Prompt-injection attempts in documents and inputs
- Requests for personal or confidential data
- Questions outside the allowed scope
- Jailbreak and role-play style attacks

ONGOING

Evaluating LLM systems after launch

KEY IDEA

Model providers update models, prompts change, and user behaviour shifts. Rerun the evaluation suite on every change, just like software tests.

PRO TIP

Collect lightweight live signals: thumbs up or down, edits users make to drafts, escalations to humans, and a regular sample of conversations reviewed against the rubric.

IN THE WILD

A support assistant passed its evaluation suite at launch. A silent upstream model update changed its tone and made it more verbose. Scheduled re-evaluation caught the change within a day.

SECTION 15 RECAP

Evaluating LLMs and generative models

REMEMBER

Takeaways

- Build a small, real, versioned evaluation set
- Write rubrics with examples for each level
- Validate any LLM judge against humans
- Evaluate retrieval and generation separately
- Red-team and re-evaluate on every change

CHECK YOURSELF

Your LLM judge rates system B much higher than A, but B's answers are twice as long. What should you check?

ANSWER

Length bias. Compare against human ratings, control for length, and check whether the extra content is actually correct and useful.

SECTION 16

16 Validation in production

The final and longest stage: proving the model works with real users, and catching the moment it stops working.

Shadow mode · Canary · A/B tests · Monitoring · Drift · Retraining

PRODUCTION

Offline validation is a forecast. Production is the result.

No held-out set fully captures real users, real systems and a changing world. Plan to keep validating after launch.

SAFE ROLLOUT

Shadow deployment

The new model receives real production traffic and makes predictions, but those predictions are only logged, not acted on. The old system keeps making the actual decisions.

KEY IDEA

Shadow mode checks that the model works on real data, through the real pipeline, at real speed, with zero risk to users.

USE IT WHEN

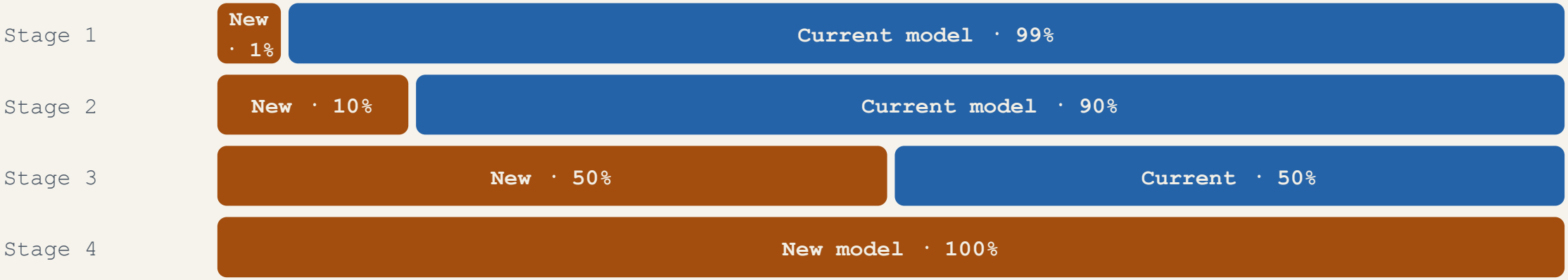
Before any user-facing launch; whenever the feature pipeline changes; when offline data may not match live data.

PITFALL

Shadow mode cannot measure effects on behaviour, because nobody acts on the predictions. You still need an online experiment for that.

SAFE ROLLOUT

Canary releases



DEFINITION

Release to a small share of traffic first, watch technical and business health checks, then expand step by step.

PRO TIP

Define automatic rollback triggers in advance: error rates, latency, or a key metric falling past an agreed limit.

CAUSAL EVIDENCE

A/B testing models

DEFINITION

Randomly assign users to the current model (control) or new model (treatment), then compare business outcomes over a pre-agreed period.

KEY IDEA

Randomisation is what lets you say the model **caused** the difference, rather than seasonality or a marketing campaign.

DO

- Pick one primary metric in advance
- Calculate the sample size needed
- Run for full weekly cycles
- Check guardrail metrics too

AVOID

- Stop the moment results look good
- Change the metric after seeing data
- Ignore novelty effects that fade

CAUSAL EVIDENCE

When A/B tests are hard

Challenge	Alternative
Network effects between users	Randomise by cluster, region or marketplace
Ranking comparisons	Interleaving: mix both models' results in one list
Many options to explore	Multi-armed bandits that shift traffic to winners
Cannot randomise for ethical reasons	Careful quasi-experiments or staged rollouts by region
Very long-term outcomes	Validated short-term proxy metrics, plus long holdouts

PRO TIP

Keep a small long-term holdout group that never gets the new model. It measures the cumulative effect of many model changes over months.

MONITORING

What to monitor after launch

Data quality

Missing values, schema changes, broken upstream feeds, out-of-range values.

Input drift

Changes in the distribution of each feature compared with training.

Prediction drift

Shifts in the distribution of scores and positive rates.

Live performance

Metrics computed once true labels arrive.

System health

Latency, error rates, throughput, cost.

Business impact

The outcome the model was built to improve.

MONITORING

Kinds of drift

DEFINITION

Data drift: the inputs change, such as a new customer segment or a sensor recalibration.

DEFINITION

Label drift: outcome rates change, such as a rise in fraud or seasonal demand.

DEFINITION

Concept drift: the relationship changes, so the same inputs now mean something different.

PRO TIP

Drift measures such as population stability or distribution distance tests are **alarms, not verdicts**. Drift in an unimportant feature may not matter; investigate before retraining.

PITFALL

Labels often arrive weeks or months later. Use input and prediction drift as early warnings until true performance can be measured.

MONITORING

Closing the loop on labels

⚠ PITFALL

When the model's decisions change what you observe (a declined loan never shows whether it would have been repaid), new labels become biased. This is a **feedback loop**.

THE FIX

Keep a small random exploration or holdout, log decision probabilities, and use human review samples to get unbiased labels over time.

IN THE WILD

A fraud model blocked most suspicious transactions, so they were never confirmed as fraud. Over time, the training data contained fewer examples of the patterns it blocked, and retrained versions grew weaker at catching them.

KEEPING IT FRESH

Retraining as a validated process

1. **Trigger**
   Scheduled, or driven by drift and performance alerts.

2. **Retrain**
   Same pipeline, newer data, versioned inputs.

3. **Validate automatically**
   Time-based test, baselines, slices, fairness and behavioural tests.

4. **Compare to champion**
   The challenger must beat the current model by an agreed margin.

5. **Roll out safely**
   Shadow or canary before full release, with rollback ready.

SECTION 16 RECAP

Validation in production

REMEMBER

Takeaways

- Shadow first: real data, zero risk
- Canary with automatic rollback triggers
- A/B tests give causal evidence of impact
- Monitor data, predictions, performance and business
- Treat retraining as a validated pipeline

CHECK YOURSELF

Prediction drift alarms fire, but labels arrive in 60 days. What can you do now?

ANSWER

Investigate which features drifted and why, review a sample manually, check business proxies, and consider a safe fallback if the shift is severe.

SECTION 17

17

Governance, documentation and MLOps

Making validation repeatable, reviewable and auditable, from model cards to automated checks in the pipeline.

Validation plan · Model cards · Independent review · Regulation · CI for ML · Reproducibility

START RIGHT

The validation plan

KEY IDEA

Write the plan **before** modelling, agree it with stakeholders, and keep it under version control. It is your protection against moving the goalposts.

PRO TIP

Treat any later change to the plan as a decision that must be justified and recorded, not a quiet edit.

- Intended use and users
- Prediction moment and inputs
- Split strategy and test set
- Primary and guardrail metrics
- Baselines to beat
- Slices and fairness criteria
- Pass and fail thresholds
- Rollout and monitoring plan

DOCUMENTATION

Model cards and datasheets

DEFINITION

Model card

A **model card** is a short, structured document describing what a model is for, how it was validated, how it performs across groups, and its known limitations.

DEFINITION

Datasheet

A **datasheet for a dataset** records how data was collected, labelled and cleaned, what it covers, and what it should not be used for.

Model card section	Contents
Intended use	Users, decisions supported, out-of-scope uses
Evaluation data	Source, time period, split method, size
Results	Metrics with intervals, overall and by slice
Limitations	Known failure modes, conditions not tested

INDEPENDENT CHALLENGE

Model risk management and review

KEY IDEA

In regulated industries such as banking, models are reviewed by an independent validation team that did not build them. They challenge assumptions, rerun tests and can block deployment.

WHY IT MATTERS

Builders are invested in their models. An independent reviewer brings fresh scepticism and catches errors the team has become blind to.

PRO TIP

Even without a formal function, ask a colleague outside the project to try to break your validation. Pay them in coffee.

WORKED EXAMPLE

Banking supervisors on both sides of the Atlantic publish model risk management expectations, covering independent validation, documentation, ongoing monitoring and model inventories.

REGULATION

The regulatory direction of travel

European Union

The EU AI Act sets requirements for high-risk AI systems, including data governance, accuracy and robustness, technical documentation, human oversight and post-market monitoring.

United Kingdom

A principles-based approach led by existing regulators, such as those for data protection, financial services and medical devices, applying principles like safety, transparency and fairness.

Sector rules

Medical devices, credit and insurance already carry specific validation and evidence expectations in many countries.

REMEMBER

This is a summary, not legal advice. Requirements and timelines evolve; check current guidance with your legal and compliance teams for your sector and region.

AUTOMATION

Continuous validation in the ML pipeline

1 → 2 → 3 → 4 → 5

Data checks

Schema, ranges, missing values, duplicates, leakage tests.

Training

Versioned data, code and configuration.

Model checks

Metrics vs baseline and champion, slices, fairness.

Behaviour tests

Invariance, direction, safety suites.

Gate

Only promote models that pass every check.

KEY IDEA

Validation should be **code**, not a notebook someone remembers to run. Every model version passes through the same automated gates.

AUTOMATION

A model quality gate in code

PYTHON · ILLUSTRATIVE

```python
def quality_gate(cand, champ, slices):
    checks = {
        "beats_baseline": cand.ap > baseline.ap + 0.02,
        "not_worse": cand.ap_low >= champ.ap - 0.01,
        "calibrated": cand.ece <= 0.03,
        "slices_ok": all(s.recall >= 0.6
                         for s in slices),
        "latency_ok": cand.p95_ms <= 120,
    }
    failed = [k for k, ok in checks.items() if not ok]
    return len(failed) == 0, failed
```

PRO TIP

Thresholds in the gate come from the validation plan, not from whatever the latest model happens to achieve.

PITFALL

The numbers here are placeholders. Set yours from business needs and measured uncertainty.

REPRODUCIBILITY

Can someone else get the same result?

- Data versioned and snapshotted, with split IDs saved
- Code and environment pinned (packages, versions)
- Random seeds recorded for splits and training
- Experiment tracking with parameters and metrics
- Evaluation scripts stored with the model
- Every test-set evaluation logged with date and author

RULE OF THUMB

If you cannot reproduce a validation result, you cannot defend it. Reproducibility is the foundation of audit and trust.

SECTION 17 RECAP

Governance, documentation and MLOps

REMEMBER

Takeaways

- Write and freeze a validation plan first
- Publish a model card with slice results and limits
- Seek independent challenge
- Automate validation as pipeline gates
- Make every result reproducible

CHECK YOURSELF

A model passed validation six months ago. A new version is ready. Which checks should run?

ANSWER

All of them, automatically: data checks, metrics against baseline and champion, slices, fairness, behaviour tests, then shadow or canary rollout.

SECTION 18

18 End-to-end case studies

Three realistic projects walked through from framing to production, showing how the pieces fit together.

Credit risk | Medical imaging | Retail demand | LLM support assistant

CASE 1 · CREDIT RISK

Framing and splitting

IN THE WILD

The brief

A lender wants to predict whether a new applicant will miss payments within 12 months, to support approval decisions.

ASK YOURSELF

When is the prediction made? At application. What is known then? Only application data and credit history up to that date.



KEY IDEA

Split by application date with a 12-month gap so every label has fully matured. Customers who reapply stay on one side of the split.

CASE 1 · CREDIT RISK

Metrics, leakage and fairness

DEFINITION

Metrics: ROC AUC for ranking, calibration for pricing, expected loss at the approval threshold for the business.

PITFALL

Leakage found: a “collections flag” field was updated after defaults. Removed after a timeline review.

KEY IDEA

Fairness: approval rates and calibration compared across age bands and other monitored groups.

WARNING SIGNS

Reject inference problem: past declined applicants have no outcome labels, so training data under-represents riskier profiles.

THE FIX

Mitigation: a small randomised approval band for borderline cases to collect unbiased labels over time.

LESSON

Independent model risk review rebuilt the test set from raw data and confirmed the results before sign-off.

CASE 2 · MEDICAL IMAGING

Validation that matches clinical use

IN THE WILD

The brief

A model flags chest X-rays likely to show a serious finding, so radiologists can review them first.

PITFALL

The first validation used random image splits. Patients with several scans appeared on both sides, and scores were inflated.

THE FIX

- Split by patient
- External test at a different hospital
- Reference labels from a panel of radiologists
- Report recall at a fixed false-alarm rate
- Slice by scanner type, age and sex

CASE 2 · MEDICAL IMAGING

What external validation revealed

WARNING SIGNS

Shortcut found

Performance dropped at the external hospital. Explanations showed attention on text markers and portable-scanner artefacts typical of very sick inpatients.

THE FIX

Response

Masked text markers, added more varied training data, and ran a prospective silent trial where predictions were logged but not shown to clinicians.

LESSON

Internal test sets share hidden quirks with training data. For high-stakes models, external and prospective validation are not optional extras; they are the real test.

CASE 3 · RETAIL DEMAND

Forecasting thousands of products

IN THE WILD

The brief

A grocer forecasts daily demand for 8,000 products across 300 stores to plan orders.

KEY IDEA

Rolling-origin evaluation over the past year, forecasting 1 to 14 days ahead, scored with a scaled error against seasonal naive.

Check	Finding
Overall vs naive	Clear improvement
New products	Worse than naive
Promotions	Big under-forecasts
Holidays	Missed peaks
Fresh produce	Strong gains

CASE 3 · RETAIL DEMAND

Turning findings into decisions

THE FIX

New products: fall back to a similar-product model until enough history exists.

THE FIX

Promotions: add planned-promotion features, built only from information known when orders are placed.

THE FIX

Business metric: simulate stock-outs and waste using the forecasts, not just error scores.

KEY IDEA

Rollout: run in shadow alongside planners for six weeks, comparing order recommendations.

PRO TIP

Monitoring: weekly scaled-error dashboards by category and store, with drift alerts.

LESSON

The overall score hid both the biggest wins and the biggest risks. Slicing drove every key decision.

CASE 4 · LLM SUPPORT ASSISTANT

Evaluating a retrieval-based assistant

IN THE WILD

The brief

A company launches an assistant that answers customer questions using its help-centre articles.

KEY IDEA

An evaluation set of 300 real, anonymised questions, including 40 with no answer in the help centre and 30 adversarial prompts.

Layer	Measure
Retrieval	Right article in top 5
Faithfulness	Claims supported by sources
Correctness	Rubric, human plus judge
Safety	Refusals, no data leaks
Live	Escalation rate, ratings

CASE 4 · LLM SUPPORT ASSISTANT

What the evaluation caught

WARNING SIGNS

Problem

When no article answered the question, the assistant often invented a plausible-sounding policy instead of saying it did not know.

THE FIX

Fix

Prompt and retrieval changes to encourage “I don’t know” with a handover, plus a dedicated test category for unanswerable questions.

LESSON

Include the questions your system **should not** answer. They are where generative systems fail most dangerously, and they are often missing from evaluation sets.

19

SECTION 19

Toolkit and cheat sheets

Checklists, quick-reference tables and a final self-test to use on every project.

Master checklist | Red flags | Report template | Tools | Glossary | Final quiz

MASTER CHECKLIST

Before training

- [ ] Use case, users and decision clearly defined
- [ ] Prediction moment and available inputs documented
- [ ] Cost of each error type agreed
- [ ] Primary metric and guardrails chosen
- [ ] Baselines defined
- [ ] Data deduplicated
- [ ] Split strategy matches deployment (group, time, site)
- [ ] Test set created and locked away
- [ ] Important slices and fairness criteria listed
- [ ] Validation plan written and versioned

MASTER CHECKLIST

Before launch

- [ ] All preprocessing fitted inside pipelines
- [ ] Tuning nested or kept off the test set
- [ ] Test set evaluated once, on the frozen model
- [ ] Results reported with intervals
- [ ] Beats baselines by a meaningful margin
- [ ] Slices, fairness and calibration checked
- [ ] Robustness and behaviour tests passed
- [ ] Top features reviewed for leakage
- [ ] Model card and validation report complete
- [ ] Shadow, canary and rollback plan ready

QUICK REFERENCE

Red flags that should stop you

⚠ PITFALL

A score far above published results or intuition.

⚠ PITFALL

One feature dominates importance.

⚠ PITFALL

Random splits on time-ordered or grouped data.

⚠ PITFALL

No baseline in the results.

⚠ PITFALL

A single number with no range.

⚠ PITFALL

Test set used many times during development.

⚠ PITFALL

Accuracy reported on heavily imbalanced data.

⚠ PITFALL

No results broken down by slice.

⚠ PITFALL

No plan for monitoring after launch.

QUICK REFERENCE

A validation report template

Section	What to include
1. Summary	Decision supported, headline result with interval, recommendation
2. Data	Sources, dates, size, labelling, known gaps
3. Method	Split design, CV scheme, tuning, threshold selection
4. Results	Primary and secondary metrics vs baselines, with intervals
5. Deep dives	Slices, fairness, calibration, robustness, error analysis
6. Limitations	Untested conditions, risks, mitigations
7. Deployment	Rollout plan, monitoring, retraining triggers, owners

QUICK REFERENCE

Useful tools

Need	Examples of tools
Splits, CV, metrics, pipelines	scikit-learn, imbalanced-learn
Experiment tracking	MLflow, Weights & Biases
Data validation	Great Expectations, Pandera, TensorFlow Data Validation
Fairness	Fairlearn, AIF360
Explainability	SHAP, scikit-learn inspection tools, Captum
Monitoring and drift	Evidently, NannyML, cloud monitoring services
LLM evaluation	Evaluation harnesses, RAG evaluation libraries, custom rubric scripts

PRO TIP

Tools change quickly. The principles in this deck do not. Pick tools that make the principles easy to follow, and check each project's current documentation.

GLOSSARY

Key terms, part 1

Term	Plain meaning
Generalisation	Performing well on new, unseen data
Overfitting	Learning noise and quirks of the training data
Hold-out set	Data kept aside for checking, not training
Cross-validation	Rotating which part of the data is held out
Stratification	Keeping class proportions equal across splits
Data leakage	Information unavailable at prediction time used in training
Nested CV	Tuning inside an outer loop for honest estimates

GLOSSARY

Key terms, part 2

Term	Plain meaning
Precision	When the model says yes, how often it is right
Recall	Of all real cases, the share the model finds
Calibration	Predicted probabilities match observed rates
Bootstrap	Resampling the test set to estimate uncertainty
Drift	Data or relationships changing over time
Shadow mode	Running a model live without acting on its output
Model card	Short document of a model’s use, results and limits

FINAL SELF-TEST

Five questions to check your understanding

1. **Splitting**
   Why is a random split wrong for predicting next month’s sales?

2. **Leakage**
   Name two ways preprocessing can leak test information into training.

3. **Metrics**
   Why can a model with 99% accuracy be useless?

4. **Uncertainty**
   What does it mean if the interval on a model difference includes zero?

5. **Production**
   What can shadow mode tell you, and what can it not?

FINAL SELF-TEST

Answers

ANSWER

1 · Splitting

Random splits let the model learn from future months. Split by time, with a gap if labels take time to mature.

ANSWER

2 · Leakage

Scaling, imputation, feature selection or target encoding fitted on the full dataset before splitting.

ANSWER

3 · Metrics

With heavy imbalance, always predicting the majority class achieves it while catching nothing.

ANSWER

4 · Uncertainty

The data does not show a real difference. Treat the models as equivalent or gather more evidence.

ANSWER

5 · Production

Shadow mode shows real-data behaviour and pipeline health. It cannot show effects on user behaviour or outcomes.

PRO TIP

Score yourself honestly. Any wrong answer points to the section to revisit.

PULLING IT TOGETHER

The ten commandments of model validation

1

Simulate deployment

Splits mirror real use.

2

Guard the test set

Look once, at the end.

3

Beat baselines

Simple first, always.

4

Match metric to cost

Errors are not equal.

5

Hunt leakage

Too good means suspicious.

6

Report ranges

Intervals, not points.

7

Slice everything

Averages hide failures.

8

Check calibration

Probabilities must mean it.

9

Document it

Plans, cards, reports.

10

Keep validating

Production is the real test.

FINAL THOUGHT

Validation is not the step that slows you down. It is the step that lets you move fast without breaking trust.

Every hour invested in honest evidence saves days of rework, rollbacks and difficult conversations later.

THANK YOU

Questions and discussion

Bring a model you are working on. We will walk through its split, metrics, leakage risks and production plan together, using the checklists in section 19.

Anthony Chinedu Echem

Machine Learning Engineer · Phexara Ltd